BY ORDER OF THE COMMANDER
AIR COMBAT COMMAND

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Flying Operations

EA-37B OPERATIONS PROCEDURES

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This publication implements Air Force Manual (AFMAN) 11-202, Volume 3, *Flight Operations*. It establishes operational guidance for all EA-37B COMPASS CALL aircraft and aircrew. This manual applies to all Regular Air Force COMPASS CALL units. This publication does not apply to the Air National Guard, Air Force Reserve, or the United States Space Force. This is a specialized publication intended for use by Airmen who have graduated from technical training related to this publication. Ensure all records generated as a result of processes prescribed in this publication adhere to Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with (IAW) the Air Force Records Disposition Schedule, located in the Air Force Records Information Management System. Personnel at all echelons are encouraged to submit changes using a Department of the Air Force (DAF) Form 847, *Recommendation for Change of Product*. Air Combat Command Airborne Reconnaissance office (ACC/A3CR) (205 Dodd Blvd., Suite 101, Langley AFB, VA 23665-2789) will forward approved recommendations to Air Force Aircrew Task Force (AF/ACTF) for final approval prior to publication. This publication may be supplemented at any level, but all supplements must be routed to the Office of Primary Responsibility (OPR) of this publication for coordination prior to certification and approval. The authorities to waive wing or unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See Department of the Air Force Manual (DAFMAN) 90-161, *Publishing Processes and Procedures*, for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the publication's OPR for non-tiered compliance items. See **paragraph 1.3** of this publication for further information on waiver authority. The use of the name

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Chapter 1

GENERAL INFORMATION

1.1. General:

1.1.1. This manual provides guidance for operating the EA-37B. It is an original source document for many areas, but for efficiency reaffirms information found in aircraft flight manuals, flight information publications (FLIP), and other Air Force (AF) directives. When guidance in this manual conflicts with another basic/source document, the basic/source document takes precedence. When this manual is the source document, waiver authority is IAW **paragraph 1.3**. For matters where this manual duplicates information from another document, follow waiver authority outlined in the basic/source document. This publication cannot address every situation that may arise during flight operations, therefore, aircrew should utilize sound judgement, experience, and available resources to safely apply the guidance contained in this regulation.

1.1.2. Unit commanders and agency directors involved with or supporting EA-37B operations will make current copies (electronic or hardcopy) of this manual available to appropriate personnel. **(T-3)**

1.2. Key Words Explained.

1.2.1. "Will", or “Must” indicate a mandatory requirement.

1.2.2. "Should" indicates a preferred, but not mandatory, method of accomplishment.

1.2.3. "May" indicates an acceptable or suggested means of accomplishment.

1.2.4. "Note" indicates operating procedures, techniques, etc., considered essential to emphasize.

1.2.5. “Caution” indicates operating procedures, techniques, etc., which could result in damage to equipment if not carefully followed.

1.2.6. “Warning” indicates operating procedures, techniques, etc., which could result in personal injury or loss of life if not carefully followed.

1.3. Deviations and Waivers . Do not deviate from policies in this manual except when the situation demands immediate action to ensure safety. The aircraft commander (AC) is vested with ultimate mission authority and is responsible for each course of action taken.

1.3.1. Deviations. The AC shall report deviations or exceptions taken without a waiver through command channels to the Air Combat Command, Command and Control, Intelligence, Surveillance, and Reconnaissance Operations Division (ACC/A3C) within 48 hours for follow-on action. **(T-2)**

1.3.2. Waivers. The authorities to waive wing/unit level requirements in this publication are identified with a Tier (“T-0, T-1, T-2, T-3”) number following the compliance statements. See DAFMAN 90-161, *Publications Processes and Procedures*, for a description of the authorities associated with the Tier numbers. Submit waiver requests through the chain of command to the appropriate tier waiver approval authority.

1.3.2.1. Tier 2 waiver authority is delegated to ACC/A3 by the ACC/CC.

1.3.2.2. Tier 3 waiver authority is delegated to the 55 Electronic Combat Group Commander (ECG/CC) by the 55 Wing Commander for this publication.

1.3.2.3. Waivers affecting theater unique circumstances without an expiration date must be approved by, or coordinated through, the Air Combat Command Director of Operations (ACC/A3). **(T-2)**

1.3.2.4. When Changed Operational Command (CHOPed) to another MAJCOM, forward waiver requests through the chain of command to MAJCOM standardization/evaluation, Combined/Joint Force Air Component Commander (C/JFACC), or Commander Air Force Forces (COMAFFOR), as appropriate, with informational copies to ACC/A3, Air Combat Command Flight Operations Division (ACC/A3T), and ACC/A3C.

1.3.3. Waiver Periods. If the waiver period is not specified in the waiver, Tier 1, 2, and 3 waivers automatically expire 90 days after the approving commander's change of command, unless renewed by the incoming commander. For waivers issued by other authorities specified in this manual, duration of the waiver should be included with the approval. ACC/A3 approves long-term (permanent) waivers. **(T-3)**

1.4. Local Supplement Coordination Process. Operations Group commanders (OG/CCs) may define operating procedures to this manual in a unit supplement or locally generated guidance. OG/CCs must obtain approval from Air Combat Command (ACC) prior to releasing their supplement or operating guidance. **(T-2)** Send an electronic copy of the approved version to 16th Air Force Standardization and Evaluations (Stan/Eval) (16 AF/A3V). 16AF/A3V will send approved copies to ACC/A3TV.

1.5. Definitions. Find explanations or definitions of terms and abbreviations commonly used in the aviation community in Title 14 Code of Federal Regulations (CFR), Part 1, General Definitions; War Department (DOW) FLIP, *General Planning*, Chapter 2; and Dictionary of Military and Associated Terms. See **Attachment 1** for common terms used in this manual.

1.6. Roles and Responsibilities.

1.6.1. Air Combat Command Directorate of Operations (ACC/A3). ACC/A3 will provide guidance and approve waivers (as required) where specified throughout this manual.

1.6.2. Detachment Commander (DETCO).

1.6.2.1. Operational units should designate a DETCO when one or more aircraft are performing missions away from home station. The DETCO will assume responsibility for mission execution, personnel supervision (aircrew and maintenance members), lodging and transportation requirements, and coordination with higher headquarters. The DETCO is the final authority responsible for ensuring aircrews have properly coordinated mission details and will ensure all personnel complete required pre- mission briefings (e.g., local area procedures, Rules of Engagement (ROE), Special Instructions (SPINS)). They will coordinate with all respective agencies that affect mission execution and will perform any additional duties directed by the squadron commander (SQ/CC) or operations officer (SQ/DO).

1.6.2.2. When applicable (e.g. contingency deployments) and outside 55WG control, DETCOs will be placed on G-series orders.

1.6.3. Aircraft Commander. The AC is a Pilot-in-Command (PIC) qualified pilot, designated by competent authority, regardless of rank, as being responsible for and is the final authority for the operation of the aircraft. The AC will ensure the aircraft is not operated in a careless, reckless, or irresponsible manner that could endanger life or property. The AC will ensure compliance with this manual and the following:

1.6.3.1. HAF, MAJCOM, and Mission Design Series (MDS)-specific guidance.

1.6.3.2. Flight Information Publications (FLIP) and Foreign Clearance Guide (FCG).

1.6.3.3. Air Traffic Control (ATC) clearances.

1.6.3.4. Notices to Airmen (NOTAMs).

1.6.3.5. Aircraft Technical Orders (T.O.s).

1.6.3.6. Combatant Commander's instructions and other associated directives.

1.6.4. Mission Commander (MC).

1.6.4.1. MCs are desired during operations involving multiple EA-37B aircraft. They are responsible for the overall mission planning process when acting as the Mission Planning Cell (MPC) Chief. The MC will function as the mission execution authority and coordinate electromagnetic attack activity on missions involving multiple EA-37B aircraft. The AC is the final authority for operation of the aircraft. MCs are not intended to serve as logistics coordinators for aircraft movement or duplicate the administrative roles normally accomplished by a commander, operations officer, DETCO or AC. However, an MC will ensure that executing agencies are updated following mission completion.

1.6.4.2. MCs will be designated through a separate process and may fulfill this role in conjunction with their primary aircrew duties.

1.6.5. Mission Crew Members. The AC maintains ultimate responsibility for the overall conduct of the sortie. However, qualified aircrew maintain responsibility for the actions they take.

1.6.6. **Mission Lead (ML).** A mission lead will be designated for each EA-37B mission sortie/event. The ML will be responsible for leading mission planning, mission briefing/debriefing, and overall supervision of tasked mission execution. AC-qualified pilots and Mission Crew Commanders (MCCs) are eligible to be designated as ML.

1.6.6.1. ML should be the AC for an aircraft mission sortie. ML will be the MCC for an unlinked CCMCS event. ML duties may be delegated by an IP/I-MCC to an AC or MCC under their supervision for the purposes of syllabus training or aircrew development.

1.6.6.2. In the event of syllabus-dictated training for an MCC is scheduled on a mission sortie, with no commensurate AC syllabus training, the ML will be the I-MCC of record.

1.6.6.2.1. For an MCC periodic MSN evaluation scheduled on a sortie with no scheduled AC MSN evaluation, the MCC will be designated as ML. In the event that an AC and MCC MSN evaluation are scheduled on the same sortie, evaluators will coordinate for their evaluatee's to sufficiently demonstrate their ability to fulfill or support the ML role.

1.6.6.2.2. SQ/DO may designate ML to another qualified member of the crew at their discretion to best meet mission or training requirements.

1.6.6.3. The ML duty does not require an additional crewmember.

1.7. Aircrew Operational Reports. The reporting requirements in this manual are exempt from licensing IAW AFI 33-324, *The Air Force Information Collections and Reports Management Program*.

1.7.1. Mission Monitoring. Enroute reports are required only when specified in an Operations Order (OPORD)/Operations Plan (OPLAN). Squadron operations centers (SOCs) and the wing command post use aircrew reports to track Continental United States (CONUS) movements for non-sensitive missions and missions not CHOPed.

1.7.2. The ACC Command Post monitors all assigned aircraft that move to, from, or between Outside the Continental United States (OCONUS) locations. Key components of the ACC Command and Control (C2) system are the Global Command and Control System (GCCS) and various C2 facilities at theater and wing locations. When aircraft are deployed in support of operations and exercises, the command center may obtain additional information with Situation Reports (SITREPs) and Deployed Status Reports (DSRs).

1.7.3. Host wing command posts relay OCONUS movement information for ACC-assigned aircraft to the ACC Command Post via telephone notification (Defense Switched Network (DSN): 574-1555; Commercial: 757-764-1555).

1.7.4. The DETCO, or AC when a DETCO is not assigned, is responsible for the timely reporting of aircraft movement and mission progress to applicable C2 or executing agencies either through direct communication or via the enroute facility's local command post. These procedures may be modified to meet local/contingency requirements. **Note:** C2 procedures for missions deemed sensitive in nature are outlined in the tasking directive.

1.7.5. Missions at Bases with a C2 Facility. The DETCO/AC will contact the destination airfield IAW the Flight Information Handbook (FIH)/Enroute Supplement approximately 30 minutes prior to landing and include the aircraft call sign, estimated time of arrival (ETA), very important person code and requirements, maintenance status, and any additional servicing requirements. After landing, the DETCO/AC will update the applicable C2 facility with ground handling requirements and departure information.

1.7.5.1. When operating OCONUS, CONUS-based crews will inform their respective SOC/command post of actual takeoff/landing times, next projected takeoff times, and any other pertinent information. Home station C2 agencies will relay aircraft information to the ACC Command Post. These actions inform the ACC commander of the location and status of OCONUS forces. When forces CHOP to a different theater commander, report aircraft information through theater C2 agencies upon arrival in the assigned area of responsibility.

1.7.5.2. Missions at Bases without a C2 Facility. DETCOs or ACs will report actual takeoff/landing times, maintenance status, projected takeoff times, and other pertinent information to the host wing command post or command/operations center as soon as possible after the event when crew duties and safety permit.

1.7.5.3. Methods of communicating this information include, but are not limited to, High Frequency (HF) phone patch, satellite communication radio, DSN, or commercial telephone.

1.7.5.3.1. If unable to contact the host wing command post or command/operations center, retain information for submission when contact is re-established. Report communication difficulties through the chain of command. Refer to FLIP, FIH, and USAF HF/Single Side Band Airways and C2 Station section for guidance on mission reporting. Upon initial contact, confirm your arrival message has been received and update your ETA.

1.7.5.3.2. If your arrival message has not been received, transmit information to the destination as necessary. Restrict HF transmissions to operational traffic, such as movement reporting, itinerary revisions, maintenance status, flight plan information, aircraft emergencies, or other important flight information.

1.8. Mission Clearance Decision. The agency with Operational Control (OPCON) and the AC are the final decision authorities to refuse/delay a mission when, in the opinion of either, conditions are not safe to start or continue a mission. However, the AC has final responsibility for the safe execution of the mission. If the decision is made to refuse/delay a mission, the aircraft will not depart until the conditions affecting this decision have been corrected or improved to allow safe mission execution. Another AC and aircrew will not be alerted to take the same mission under the same conditions.

1.9. Aircraft Divert or Rerouting. Mission changes must be authorized by the commander with OPCON. The controlling agency directing the changes is responsible for ensuring destination requirements and facilities are adequate for the aircraft and aircrew. However, it is the AC's responsibility to ensure all changes allow for safe operation and bed down of the aircraft. The AC will notify the controlling agency of any limitations that may impede the directed changes. **(T-3)** Reference **Chapter 6** of this volume for specific aircraft security requirements.

Chapter 2

CREW COMPLEMENT

2.1. Aircrew Qualification. Primary aircrew members, or those occupying a primary position during flight, will be qualified or in qualification training for that crew position, mission, and aircraft. If non-current or in training for a particular event, the aircrew member must be under the supervision of an instructor while accomplishing that event (direct supervision for critical phases of flight, as defined in **Attachment 1**). **(T-3)**

2.2. Crew Complement.

2.2.1. Minimum basic crew is defined as one Aircraft Commander (AC) and one additional EA-37B qualified pilot. **(T-3)**

2.2.1.1. Any EA-37B qualified pilot may occupy either pilot seat.

2.2.1.2. During critical phases of flight a PIC-qualified pilot must occupy one of the pilot seats.

2.2.1.3. A PIC-qualified pilot is defined as a current and qualified EA-37B pilot who has graduated from an AC qualification syllabus. PIC-qualified denotes a pilot's level of training, but is not intrinsically tied to their current crew position or qualification as an AC. See ACCMAN 11-2EA-37B, Volume 1, *EA-37B Aircrew Training*, for AC qualification requirements.

2.2.1.4. During critical phases of flight when an unqualified pilot occupies one of the pilot seats, they will be under the direct supervision of a qualified EA-37B Instructor Pilot (IP).

2.2.2. Minimum mission crew is defined as basic crew with one Airborne Maintenance Technician (AMT) and one MCC.

2.2.3. Augmented Crew is not applicable to the EA-37B.

2.2.4. A Mission Commander (MC) is desired when conducting an operation with multiple EA-37B aircraft. An additional crew member is not required when the MC occupies their primary crew duty position (AC, MCC).

2.2.5. Senior leaders who complete a Senior Staff Qualification course (restricted AF Form 8) or orientation for a Senior Staff Familiarization flight may occupy a primary crew position when under direct instructor supervision. For additional information reference DAFMAN 11-401, *Aviation Management*, Chapter 10. **Note:** Supervisory Operational Flying differs from Indoctrination Flying.

2.2.6. Crew members who complete the Senior Staff Qualification Course will log basic qualified pilot (FP) for Flight Authorization Duty Code on the Aircraft Forms.

2.2.7. Crew members who complete a Senior Staff Familiarization flight will log Observer Pilot (OP)/Observer Electronic Warfare Officer (OE) for Flight Authorization Duty Code on the Aircraft Forms.

2.3. Mission Essential Personnel (MEP). Eligibility and authority for granting MEP status is specified in DAFMAN 11-401. **Note:** Touch and go landings may be performed with MEP onboard.

2.4. Orientation, Incentive and Familiarization Flights. Orientation, Incentive and Familiarization flights will be flown IAW DAFMAN 11-401 and the MAJCOM supplement. The AC will ensure that academic information covering aircraft egress and aircrew safety equipment is briefed prior to stepping to the aircraft.

2.5. Passenger Restrictions. The EA-37B will not be utilized for passenger transport. **(T-2)**

2.6. Flight Duty Period (FDP). For aircrew members performing other duties prior to flight-related duties, FDP begins when commencing official duties. Waiver authority is IAW AFMAN 11-202V3 and this paragraph.

2.6.1. Basic FDP is 16 hours. Do not conduct multiple practice approaches/landings, FCFs, or OCFs after 12 hours FDP. Mission crew members past 12-hour FDP do not restrict performance of basic crew training. If the autopilot fails, the AC should exercise judgment in coordination with the crew to determine the best course of action, considering mission requirements, crew fatigue, remaining transit time/altitude to the landing point, inoperable Emergency Descent Mode capability, weather and other applicable factors. Waiver authority for contingencies is C/JFACC or MAJCOM/A3 of the agency with OPCON. FDP for aircraft preflights is the same as a flight crew.

2.6.2. FDP for crews with access to GO Pills is 20 hours. If the autopilot fails, the AC should exercise judgment in coordination with the crew to determine the best course of action, considering mission requirements, crew fatigue, remaining transit time/altitude to landing point, inoperable Emergency Descent Mode capability, weather and other applicable factors. Waiver authority for contingencies is C/JFACC or MAJCOM/A3 of the agency with OPCON.

2.6.3. FDP for flight examiners administering flight evaluations will not exceed 16 hours.

2.6.4. FDP may be extended IAW AFMAN11-202V3. MAJCOM/A3 or equivalent for the agency with OPCON of the aircraft is waiver authority for maximum FDP. Coordinate with C2 agencies so that follow on missions are not adversely affected.

2.6.5. FDP may be limited for crews in Individual Protective Equipment (IPE) such as the Joint Service Aircrew Mask (JSAM) due to heightened Risk Management (RM) concerns. Aircrew members in IPE may experience a degradation in flight duty performance due to increased fatigue and restrictions in situational awareness. ACs should work closely with C2 personnel to ensure an appropriate FDP is established when mission segments require IPE. See **Chapter 8** for additional guidance.

2.7. Crew Rest. Crew rest policy is IAW AFMAN11-202V3 and this paragraph.

2.7.1. Home-Station Pre-Departure. For operational missions (e.g. mobilization) scheduled away from home station for more than 24 hours, all primary aircrew members will be allotted 24 hours prior to show time. The first 12 hours are designed to allow crew members time to resolve personal affairs and base out-processing. The ECG/CC, or equivalent, is the waiver authority for the first 12 hours of pre-departure crew rest if flight related duties are required during this period, to include mission planning. The second 12 hours are considered crew rest. Aircrew members will not be manifested as passengers to reduce or eliminate crew rest requirements.

2.7.2. Enroute Ground Time and Crew Rest:

2.7.2.1. Minimum planned ground time is 16 hours between engine shutdown and mission takeoff, unless extended post-flight duties are anticipated. These duties include, but are not limited to, refueling, securing classified material, performing maintenance, or completing mission debriefings.

2.7.2.2. Crews re-enter crew rest if their aircraft or mission is not capable of departure within 4 hours of scheduled takeoff time. The SQ/DO or above may make exceptions but require the concurrence of the AC. Refer to ACCMAN11-2EA-37BV1 for additional restrictions on training missions.

2.7.3. Post-Mission Crew Rest (PMCR). SQ/CCs will grant aircrew members returning to home base sufficient time to recover from cumulative effects of the mission and tend to personal needs. **(T-3)** PMCR begins upon mission termination.

2.7.3.1. The ECG/CC (or equivalent) is the PMCR waiver authority.

2.7.3.2. For missions that keep an aircrew off station 24 or more hours, the SQ/CC will provide 1 hour (up to 24 hours) PMCR for each 3 hours off-station. **(T-3)** SQ/CCs may extend PMCR at their discretion. Do not enter aircrew members in crew rest until the PMCR period expires.

2.7.3.3. PMCR is not applicable to continuing missions.

2.7.3.4. PMCR is not applicable to training sorties or off-station trainers.

2.8. Standby Force Procedures. Note: Contingency operations may require modification of the following Standby Force Procedures. The SQ/CC or SQ/DO will approve any modification of these procedures. **(T-3)**

2.8.1. Crew Management. Except as noted below, commanders will not use a standby crew to perform any non-mission duties or duties not related to their standby status. **(T-3)** Standby crews will not preflight any aircraft other than their standby aircraft. **(T-3)**

2.8.2. ALPHA Standby Force:

2.8.2.1. An aircraft and aircrew capable of taking off in 1 hour. Aircrew members are given 12 hours of pre-standby crew rest before or after aircraft preflight. Aircrews must complete all preflight duties within 6 hours of crew show time. **(T-3)** An additional 12-hour pre-standby crew rest is required when preflight time exceeds 6 hours and crew rest was given before the preflight. Once an ALPHA force is formed, additional pre-flights may be necessary to maintain the ALPHA aircraft. Additional pre-flights done during normal waking hours do not interrupt crew rest. A crew will not stay on ALPHA standby duty for more than 48 hours. **(T-3)** After 48 hours, the crew must takeoff, be released, or be entered into pre- departure crew rest. **(T-3)** FDP begins at the time of alert.

2.8.2.2. Aircraft Security. Each unit will complete a maintenance and aircrew preflight inspection when they put an aircraft on ALPHA standby status. **(T-3)** The ALPHA Standby AC will ensure the aircraft is sealed after preflight. **(T-3)** Secure all hatches and doors to prevent unauthorized entry. After aircraft sealment, the ALPHA AC or designated representative will maintain positive control of the aircraft keys. The SQ/CC, DETCO, or designated representative must grant permission for persons, other than the ALPHA Standby crew, to enter an aircraft once the plane is secured. **(T-3)** The ALPHA Standby

AC or designated representative must be present if access to the assigned aircraft is required. **(T-3)** Resecure the standby aircraft any time the aircraft is opened, if required.

2.8.3. BRAVO Standby Force. An aircraft or aircrew capable of taking off in 3 hours from the time the crew is told to takeoff or alerted. Aircrew members are given 12 hours of pre-standby crew rest. Crews are legal for alert after pre-standby crew rest. Preflight duties, if required, interrupt crew rest. A crew will not stay on BRAVO standby duty for more than 48 hours. After 48 hours, the crew must takeoff, be released, or be entered into pre-departure crew rest. FDP begins when the crew shows for duty. If a BRAVO standby crew is alerted for any duty (takeoff, preflight, mission planning), and the unit is subsequently tasked to launch the mission, FDP is calculated from when the crew first reported for that duty.

2.8.3.1. MODIFIED BRAVO Standby Force. While in garrison, all BRAVO Standby Force criteria will remain unchanged with exception to a revised alert timeline. A recall will be initiated to alert aircrews for contingency operations or HHQ taskings. Upon alert notification, crews will have 1 hour to report to the specified location and will be prepared to takeoff within 4 hours from the initial alert notification. The ECG/CC, or designated representative, may adjust this timeline as necessary to align with mission requirements. FDP is calculated from when the crew first reports to the specified duty location.

2.8.3.2. CHARLIE Standby Force. An identified aircrew capable of entering crew rest within 2 hours after their controlling unit is notified. This aircrew becomes legal for alert 12 hours after entering crew rest. CHARLIE Standby will not exceed 72 hours. After 72 hours, the crew will be released. **(T-3)** Afford a minimum of 12 hours before resuming CHARLIE Standby duty, entering crew rest for a mission, or entering pre-standby crew rest for ALPHA or BRAVO Standby.

2.9. Counter-Fatigue Management Program.

2.9.1. Aircrew may use medications with prior approval (on a voluntary basis following ground testing) that enhance natural rest during off-cycle crew rest periods. This section provides guidance for the use of no-go medications (prescription medications) that help aircrew initiate and maintain restful sleep during off-cycle (desynchronosis) crew rest periods. Aircrew will not use no-go pills in flight. **(T-2)**

2.9.2. It is USAF policy that aircrew will never use no-go pills as a first-choice counter-fatigue management tool. **(T-2)**

2.9.3. Responsibility for counter-fatigue management of aircrew medicinal products rests with the home station flight surgeon (FS), WG/CC (may delegate to but no lower than SQ/CC), and with each individual aircrew member. During deployments, aircrew members only obtain no-go pills from an FS after review with a USAF FS.

2.9.4. Unit ORM programs include use of no-go medication with WG/CC and FS oversight.

2.9.5. The ECG/CC will establish a system to inform the FS when missions fall into any of the following categories (may cause sleep disruptions and are therefore candidates for no-go medications):

2.9.5.1. Home station night launch missions greater than four hours duration. **(T-3)**

2.9.5.2. Crew rest facilities lacking an optimal sleeping environment (quiet, climate controlled and darkened). **(T-3)**

2.9.5.3. Off-station missions that are four or more time zones from home station. **(T-3)**

2.9.5.4. Rotating schedules (stair-stepped flying schedules) with greater than 6-hour flight time duration. **(T-3)**

2.9.5.5. Missions that run 14-hour (or greater) duty day for 3 days or more. **(T-3)**

2.9.5.6. Crew members will not consume a no-go pill on a timeline where they would be under the effect of the medication while they perform aircrew duties or are on ALPHA/BRAVO status (use mission report or legal for alert time to determine latest time to take no-go medication). Adhere to guidance below, unless superseded by updated MAJCOM or HAF direction.

2.9.5.6.1. Temazepam (Restoril). A dose of 15-30 mg is to be taken with a minimum DOWN period of 12 hours before resuming duties.

2.9.5.6.2. Zolpidem (Ambien). A dose of 5-10 mg is to be taken with a minimum DOWN period of 6 hours before resuming duties.

2.9.5.6.3. Zaleplon (Sonata). A dose of 10 mg is to be taken with a minimum DOWN period of 4 hours before resuming duties.

2.9.5.7. Aircrew member's responsibilities:

2.9.5.7.1. Complete ground testing for no-go pills and receive FS clearance prior to using no-go pills in the operational environment.

2.9.5.7.2. Do not operate equipment within the DOWN periods for each of the no-go pills as specified in **paragraph 2.9.5.6**.

2.9.5.7.3. Do not take no-go-pills within 12 hours of consuming alcohol.

2.9.5.7.4. Inform the FS of any other medications (including nutritional supplements and over the counter medications) they are taking so the FS can evaluate potential interactions.

2.9.5.7.5. Limit use of Restoril and Ambien to a maximum of seven consecutive days and no more than 20 days in a 60-day period.

2.9.5.7.6. Limit use of Sonata to a maximum of 10 consecutive days and no more than 28 days in a 60-day period.

Chapter 3

AIRCRAFT OPERATING RESTRICTIONS

3.1. Objective. This chapter augments the Gulfstream G550 Master Minimum Equipment List (MMEL) with EA-37B specific modifications and applies to accepting an aircraft from maintenance prior to takeoff. The ultimate objective of the aircraft maintenance team is to provide an aircraft ready for flight with all equipment operational. Manpower limitations, skills, and spare part availability may have a negative and direct impact on mission accomplishment. However, under specific circumstances, some missions can be safely operated without all equipment being operational. Using the following policies, the AC is the final authority in determining the overall suitability of an aircraft.

3.2. Master Minimum Equipment List (MMEL) Policy. Crews will utilize Gulfstream G550 Master Minimum Equipment List (MMEL). For repair category descriptions, utilize PL-25 in the Policy Letter tab located in the MMEL. This chapter is not intended to allow for continued operation of the aircraft for an indefinite period with systems/subsystems inoperative.

3.2.1. The AC is responsible for exercising the necessary judgment to ensure no aircraft is flown with multiple items inoperative that may result in an unsafe degradation and/or an undue increase in crew workload.

3.2.1.1. The AC should also consider the possibility of additional failures during continued operation with inoperative systems or components.

3.2.1.2. ACs should be aware that the MMEL is an operational tool and serves a different purpose than the maintenance Minimum Essential Subsystem List (MESL). The MMEL does not direct deviation from the aircraft flight manual limitations, emergency procedures, or USAF/ACC directives.

3.2.2. If, after exploring all options, the AC determines a safe takeoff is possible with a required MMEL item inoperable (beyond a particular restriction), the AC will request a waiver. **(T-3)** Plan a minimum 12-hour response to the waiver request.

3.2.2.1. If an item is needed to be operational IAW the maintenance MESL, the AC may still opt to accept the aircraft without waiver. Safety of flight is paramount.

3.2.2.2. All emergency equipment will be installed and fully operational unless not required in support of mission requirements. **(T-3)**

3.3. Waiver Protocol. Waivers to operate with degraded or inoperative equipment beyond the guidance in MMEL may be granted on a case-by-case basis and only in exceptional circumstances. Waiver authority is the ECG/CC, or equivalent, based on who has OPCON and execution of the aircraft performing a specific mission. The AC determines the need for a waiver and initiates the request.

3.4. Technical Assistance Service . The AC may (at any time in the decision process) request technical support and additional assistance from their home unit, ACC staff, and/or maintenance representatives.

3.4.1. ACs electing to operate with degraded equipment or aircraft systems (with appropriate waiver, if necessary) must coordinate mission requirements (revised departure times, fuel requirements, maintenance requirements, etc.) with the C2 agency before flight. **(T-3)**

3.4.2. If beyond C2 communication capability, the AC may deviate from this chapter or the MEL IAW **paragraph 1.3**. Report deviations (without waiver) through channels to ACC/A3C within 48 hours. Units must be prepared to collect background information and submit a follow-up written report upon request. **(T-3)**

3.5. Navigation Systems.

3.5.1. For flights in Minimum Navigation Performance Specification, refer to G550 AFM, FLIP General Planning (GP) and Area Planning (AP) volume appropriate for route of flight for equipment requirements.

3.5.2. For flights on all other Category I routes, the AC determines the minimum navigational capability required to safely accomplish the mission. Consider the following: duration, route of flight, weather, experience, and proficiency of the crew.

3.6. Equipment/Cargo Loading.

3.6.1. Due to the aircraft and equipment configuration, EA-37B aircraft are not capable of carrying cargo. Cargo is defined as any item loaded aboard the aircraft except crew baggage, professional gear, spare mission equipment, aircraft maintenance toolboxes, aircraft tow-bar, ladder or safety/emergency equipment.

3.6.2. Total weight of individual baggage will be IAW weight and balance restrictions.

3.7. Gear-Down Flight Operations.

3.7.1. Limit gear down flight operations to sorties required to move the aircraft to a suitable repair facility. Consider gear down flight only after the AC exhausts all avenues to repair the aircraft in place.

3.7.2. Standard climb-out flight path charts are in G550 OM. For gear down operations, drag index must be applied using the Twin Engine Climb, Landing Gear Extended charts. ACs will not takeoff without reasonable assurance the aircraft can achieve/maintain adequate obstacle clearance to include enroute stops and alternates.

3.8. Definitions (Specific to this Chapter).

3.8.1. Home Station. Home bases of assignment for EA-37B aircraft. Aircraft will not depart their home stations unless MMEL home station requirements are met. **Exception:** During wartime, enroute criteria applies to all aircraft departures.

3.8.2. Enroute. Enroute locations where maintenance repair capability exists. An enroute station has the necessary skilled USAF or USAF-contract maintenance personnel, support equipment, and technical data available to accomplish most repairs.

3.8.3. Local Training. A mission scheduled to originate and terminate at home station, generated for training or evaluation and executed at the local level.

3.8.4. Off-Station Training. A mission that departs home station to perform training, as directed by the wing or ECG/CC, without returning the same day. These missions are

supported by home station logistics. **Note:** Off-station trainers are considered local training for the purposes of this chapter.

3.8.5. Remarks/Limitations/Exceptions. Some technical information and procedures are contained in this column. This is not all inclusive. Crew members should refer to the flight manual and other directives for procedures, techniques, limitations, etc.

3.8.5.1. One-time Flight Clarification:

3.8.5.1.1. A Not Mission Capable (NMC) discrepancy may be downgraded for a one-time flight. This condition does not preclude carrying cargo, MEPs or additional crewmembers. The priority is to move the airplane to a repair-capable facility. ACs must coordinate with appropriate agencies to ensure repair capability exists at the destination. **(T-3)** One-time flights may include enroute stops only when necessary to recover the airplane. **Example:** An airplane departs on a gear-down flight from Djibouti initial approach point and requires an enroute fuel stop (Cairo) before landing at the nearest repair capable facility, Sigonella Naval Air Station.

3.8.5.1.2. One-time flight to nearest repair-capable facility: Flight is limited to the shortest enroute time to the nearest repair-capable base.

3.8.5.2. Other Mission and Repair Clarifications:

3.8.5.2.1. Repaired at next repair-capable facility: Mission may continue as scheduled; item is repaired upon reaching a repair-capable facility. Once maintenance action is initiated, and it is determined repairs are not possible, the AC discusses possible courses of action with C2 agency to return aircraft to service. Flights should only be conducted with intent to take the aircraft to a repair facility, not to conduct training.

3.8.5.2.2. Mission dictates requirement: AC considers the entire mission profile, not just the next leg. **Example:** An airplane is departing an enroute station with repair capability, after engine start they discover the #1 engine anti-ice is inoperative. Icing conditions are not forecasted for the next leg. However, because the mission spans several days and repair capability does not exist at the scheduled enroute stops, the AC elects to have the item repaired prior to departing.

Chapter 4

OPERATIONAL PROCEDURES

4.1. Checklists. A checklist is not complete until all items have been accomplished and all applicable crewmembers have called it complete. Momentary hesitations for coordination items, ATC interruptions, and deviations or deferred items specified in the flight manual, etc., are authorized. Notes amplifying checklist procedures or limitations may be added to the checklists in pencil.

4.1.1. The pilot flying (PF) the aircraft initiates all checklists unless another procedure is established by the flight manual or this volume. The pilot monitoring (PM) executes the checklist and ensures completion.

4.1.2. The only pages (or inserts) authorized in checklists are Gulfstream AFM/ Airplane Operating Manual (AOM) aircrew checklists, ACCMAN MDS Volume 3 or MAJCOM checklists. No unapproved items will be inserted between the page covers of authorized checklists. Authorized entries will be current and written in pencil. **(T-3)**

4.1.3. Local in-flight guides and inserts not affecting AFM/AOM guidance and procedures may be locally developed with 55 ECG/EGV approval.

4.1.4. Abbreviated checklist items that do not apply to the unit's aircraft or mission may be lined out. Do not challenge these items during checklist accomplishment.

4.1.5. Unit developed checklists (e.g., fanfold, local checklists, etc.), to include performance data inserts, may be used in lieu of flight crew checklists if they contain, as a minimum, all items from the flight crew checklists. These checklists will reflect the same change number, change date and applicable supplement numbers as the flight crew checklist for configuration control.

4.1.6. Unit-developed aircrew aids should include the following items:

4.1.6.1. Briefing guides.

4.1.6.2. Local ultra-high frequency and very high frequency channelization.

4.1.6.3. Appropriate airfield diagrams.

4.1.6.4. Emergency information (for example, emergency action checklists, no-radio procedures, divert information).

4.1.6.5. Cross-country procedures to include command and control, engine documentation, and aircraft servicing.

4.1.6.6. Other information as deemed necessary by the unit (for example, turnaround procedures, and instrument preflight).

4.1.6.7. Training rules.

4.1.6.8. Need to know local telephone numbers.

4.1.6.9. Local training areas.

4.1.6.10. Divert information.

4.2. Duty Station. A qualified pilot will be in control of the aircraft at all times during flight. **Exception:** Unqualified pilots undergoing qualification training. Only one pilot may be absent from their duty station at a time. With both pilots in their seats, ACs may authorize rest periods for one pilot occupying a primary duty station during non-critical phases of flight (the other pilot will be awake and alert). All crew members will notify the PF prior to disconnecting from communications.

4.2.1. All crew members will be at their duty stations during all critical phases of flight. **(T-3)**. Crew members may occupy other stations, with MCC or AC concurrence, only if doing so does not interfere with normal crew duties.

4.2.2. During critical phases of flight, a current and qualified MCC, or MCC under the supervision of an instructor, will occupy Workstation 5.

4.2.3. Unless crew duties dictate otherwise, an MCC will occupy Workstation 5 and an AMT will occupy Workstation 6 during mission system operation.

4.3. Flight Station Entry. ACs may authorize observers, MEP, and any crewmember access to the cockpit. Observers, MEPs and non-pilot crewmembers will not occupy the pilot or copilot positions at any time during aircraft operations.

4.4. Seat Belts.

4.4.1. All occupants will have a designated seat with a seat belt. **(T-3)**

4.4.2. All occupants will be seated with seat belts and shoulder harnesses fastened during taxi, takeoff and landing. **(T-3)**

4.4.3. Crew members occupying their designated positions will have seat belts fastened at all times, except when crew duties require otherwise. **(T-3)**

4.5. Takeoff and Landing Policy .

4.5.1. A PIC-qualified pilot will occupy either the left or right seat during all takeoffs and landings. The AC is not required to occupy a primary crew position, but still retains overall authority.

4.5.2. Non-PIC-qualified pilots may perform takeoffs and landings from either seat.

4.5.3. A PIC-qualified pilot makes all takeoffs and landings during aircraft emergencies unless conditions prevent compliance. **Exception:** When an instructor pilot (IP) is in either pilot seat, they may elect to allow a non-PIC-qualified pilot to land.

4.6. Landing Gear and Flap Operating Guidance . The PF will command configuration changes. The pilot monitoring (PM) the aircraft will verify appropriate limitations and acknowledge the command by repeating it. The landing gear will be actuated by the right-seat pilot and the flaps will normally be operated by the PM. During ground operations, the copilot may actuate the flaps without notifying the pilot. Note: When splitting preflight duties, coordinate flap movement to ensure safety of personnel outside the aircraft.

4.7. Aircraft Lighting.

4.7.1. Operate aircraft lighting IAW Gulfstream G550 Airplane Flight Manual (AFM), AFMAN 11-202V3 and AFMAN 11-218, *Aircraft Operations and Movement on the Ground.*

4.7.2. Contingency operations may dictate that external lights be off and internal lights be limited to the minimum necessary for aircrew activities.

4.8. Wake Turbulence Avoidance. The PF will comply with takeoff and landing intervals and wake turbulence separation criteria IAW AFMAN 11-202V3 and FLIP GP when receiving instructions to follow an aircraft or when receiving a visual approach clearance.

4.9. Communications Guidance.

4.9.1. Aircraft Interphone.

4.9.1.1. Do not discuss classified information over Flight Deck (FD) Net when radio transmissions are being made unless absolutely necessary for mission accomplishment.

4.9.1.2. The PF will notify the crew which interphone is primary and update the crew prior to primary interphone changes. **(T-3)** PF, PM and MCC will monitor primary interphone net at all times. **(T-3)** When available, all crew members will monitor primary interphone during ground operations and in the terminal environment. **(T-3)** Advise the PF when off interphone. During other phases of flight, mission crew members will monitor primary interphone and flight crew hot mic as directed by the PF or MCC. **(T-3)**

4.9.1.3. After engine shutdown, crew members will remain on headset until the PF clears the crew off headset. **(T-3)** Loss of Digital Intercommunication System (D-ICS) communications during quiet engine shutdown is considered clearance off headset from the PF.

4.9.1.4. Sterile Cockpit. Limit conversation to that essential for crew coordination and mission accomplishment during taxi, takeoff, approach, landing and any flight below 10,000 ft MSL.

4.9.1.5. CALL. PF will notify crew of emergencies using CALL at earliest convenience.

4.9.2. Radios.

4.9.2.1. Before the completion of the Climb Checklist and after initiating the Descent Checklists, all aircrew members will monitor the primary radio. **(T-3)**

4.9.2.2. The pilot operating the radios will notify the crew which radio is primary and update the crew when the primary radio changes. **(T-3)**

4.9.2.3. A pilot will monitor a guard frequency.

4.10. Advisory Calls.

4.10.1. The PF will announce intentions during departures, arrivals, approaches, and when circumstances require deviating from normal procedures. **(T-3)** The PM will make mandatory advisory calls except those designated for another crew member. **(T-3)**

4.10.1.1. Takeoff, Approach and Landing. Refer to G550 AOM and ETM-EA37B-QRCL-1 Climb and Approach Checklists for required callouts. **(T-3)**

4.10.1.2. In addition to callouts prescribed in **paragraph 4.10.1.**, PM will call 100 feet above all stepdowns and final approach fix (FAF) altitude with “100 Above”. This callout will be responded to by the PF with “100 Above” as a confirmation.

4.10.2. Altitudes.

4.10.2.1. Whenever a new altitude clearance is assigned by ATC and with autopilot engaged, the PF will make all changes on the guidance panel, followed by verification from the PM unless otherwise delegated by the PF.

4.10.2.2. Determine Derived Decision Altitude (DDA) IAW G550 AFM. In no case will DDA be less than 60 ft above published Minimum Descent Altitude (MDA) for a non-precision approach.

4.10.3. Touch and Go. If a malfunction is encountered, crew members should only state the observed malfunction. The PF announces their intention to stop or continue the takeoff.

4.11. Deviations .. Any crewmember will tell the PF when deviations of heading +/- 5 degrees, altitude +/- 100 ft or airspeed +/- 10 knots are observed, and no attempt is being made to correct the deviation.

4.12. Cockpit/Crew Resource Management (CRM) & Threat and Error Management (TEM). The goal of CRM/ TEM is enhancing mission effectiveness and is the responsibility of all crew members.

4.12.1. All crew members are responsible for fostering an atmosphere of open communication and crew participation in the decision-making process. Crew member responsibilities include understanding the authority of the AC and participating in the mission and decision-making process. Crew members are expected to assert their best judgment and, when in doubt, speak out. The AC is the final decision authority.

4.12.2. "Time Out" is the common assertive statement for use by all crew members when any developing situation or loss of situational awareness may jeopardize mission success. As soon as possible after a "Time Out" call, the PF will stabilize the aircraft and the MCC will bring the mission system out of Jam (as applicable). Once the aircraft is in a safe configuration, the AC will allow the crew to voice concerns. Relying on crew inputs, the AC will decide whether to continue the current course of action or pursue another.

4.12.3. "Knock It Off" is used to terminate a maneuver or scenario for safety. Upon hearing "Knock-It-Off" the crew should establish a safe attitude, altitude and airspeed, return the aircraft power and flight controls to a normal configuration, and bring the mission system out of jam. Reference AFMAN 11-214, *Air Operations Rules and Procedures,* for further details.

4.12.4. The Two Challenge Rule is used if the PM is unable to gain a response from the PF. After two challenges, if an immediate correction is not made by the PF, the PM takes control of the aircraft.

4.13. Use of Automation.

4.13.1. General Automation Procedures. Aircrews should use aircraft automation consistent with changing flight environments and aircraft capabilities. If the use of automation creates a loss of situational awareness or results in task saturation, shift to a less demanding level or disconnect the automation entirely and re-establish desired aircraft path and control. Both pilots are responsible for ensuring the aircraft is following the desired flight path.

4.13.2. Verbalize, Verify, and Monitor (VVM) is a closed-loop system of communication designed to significantly reduce typical automation selection errors between the PF and the PM. VVM consists of the following three step process:

4.13.2.1. VERBALIZE intentions prior to acting on them.

4.13.2.2. VERIFY intended actions have been made.

4.13.2.3. MONITOR the actions to ensure intended outcome(s) have occurred.

4.14. Radar Altimeter.

4.14.1. Any crew member detecting a yellow “DH” indication on the PFDs at or below the altitude set in the radar altimeter will immediately notify the PF. **(T-3)** Verify terrain clearance and aircraft position.

4.14.2. Before departure, set the radar altimeter should be set to 1500 for takeoff emergencies.

4.14.3. The pilots will use the same radar altimeter setting unless briefed otherwise. **(T-3)**

4.14.4. Set the barometric altimeter to the DA/MDA/Circling Altitude during instrument approaches.

4.15. Foreign Object Damage (FOD). To reduce the risk of FOD during ground operations, personnel will:

4.15.1. Avoid engine hazard areas.

4.15.2. Avoid using excessive power.

4.15.3. Avoid prop or jet blast from other aircraft.

4.15.4. Ensure loose items are secure in the aircraft before opening any doors.

4.16. Loose Objects in the Flight Deck. Aircrew will not place items (checklists, charts, etc.) on the throttle quadrant, trim wheel, Fire Handles, or Fuel Control switches.

4.17. In-Flight Meals. Pilots should not eat meals at the same time and their meals should consist of different menu items. Do not activate or handle Meal Ready to Eat (MRE) Flameless Ration Heaters inside the aircraft.

4.18. Ground Visual Signals. Normally, pilot and ground crew will communicate by the intercom system during all Starting Engines checks. Use the intercom system to the maximum extent possible anytime maintenance technicians are performing "red balls" on the aircraft. The pilot will not activate any system that could pose any danger to the ground crew prior to receiving proper acknowledgment from ground personnel. When not using ground intercom, visual signals will be in accordance with AFMAN 11-218. The crew chief will repeat the given signal when it is safe to operate the system.

4.19. Taxi.

4.19.1. Ground and Taxi Operations. Conduct ground and taxi operations IAW AFMAN 11-218.

4.19.2. Taxi Interval. Pilots should maintain a minimum of 100 feet when in trail of another aircraft. Operational considerations may require a shorter taxi interval, pilots will maintain vigilance during congested operations and use caution when in trail of heavy, multiengine aircraft. Use normal wingtip spacing with caution when approaching or holding for the active runway.

4.20. Runway Condition Reading (RCR) and Runway Surface Condition (RSC) Limitations .

4.20.1. When no reported RCR is available, consider the runway surface wet when water on the runway causes a reflective glare. The minimum RCR on any portion of the runway for takeoff or landing is 05.

4.20.2. Limit EA-37B aircraft operations into and out of slush- or water-covered runways. Reference G550 AFM/AOM for appropriate procedures when operating in inclement conditions.

4.20.3. The PF will ensure applicable performance data is referenced when operating on wet or contaminated runways. These conditions require a correction to effective runway length to ensure a screen height of 35 feet is met on takeoff with an engine failure.

4.21. Runway and Taxiway Requirements .

4.21.1. Minimum runway width is 75 ft/23 meters. Minimum taxiway width is 25 ft/8 meters. Minimum taxiway width for 90 degree turns with fillets is a 35-foot taxiway to a 35-foot taxiway. For 90 degree turns to/from taxiways less than 35 feet the other taxiway must be 55 feet or wider.

4.21.2. Runway Length for Landing. Minimum runway for landing is based off predicted conditions IAW the G550 Performance Handbook (PH). Consideration should be given to required furl loads on subsequent sorties. (5,000 feet for planning purposes. 7,000 feet for partial flap touch n go's)

4.21.3. Runway Length for Takeoff. Minimum runway for a normal takeoff is based off predicted conditions IAW the G550 PH. Reference Accelerate-Go Distance or Reference Accelerate-Stop Distance, whichever is greater.

4.21.4. The decision to make intersection takeoffs belongs to the PF. Takeoff and Landing Data (TOLD) computations are based on the runway remaining from the point at which the takeoff is initiated.

4.22. Aircraft Taxi and Taxi Obstruction Clearance Criteria.

4.22.1. Without wing walkers, avoid taxi obstructions by at least 25 ft; with wing walkers, by at least 10 ft.

4.22.2. When taxi clearance is doubtful, use wing walker(s). If wing walker(s) are not available, deplane aircrew member(s) to maintain obstruction clearance.

4.22.3. Arresting Cables.

4.22.3.1. Do not takeoff or land over a raised barrier. Plan to takeoff before the barrier and land beyond the barrier. **(T-3)**

4.22.3.2. Gulfstream has no analytical or test data regarding operation of the aircraft when crossing cables. Operations over raised barrier cables may result in damage to the aircraft. Refer to Engineering Coordination Memorandum L3-GAC-CD-10102, *EC-37B Operations Over Barriers*, for more information.

4.23. Bird/Wildlife Aircraft Strike Hazard (BASH) Programs. BASH programs are centralized unit efforts that provide information cross-feed, hazard identification, and a

consolidated course of action. As a minimum, squadron commanders or equivalent must implement the following procedures:

4.23.1. Ensure compliance with the following Bird Watch Condition (BWC) restrictions.

4.23.1.1. BWC Low – No operating restrictions.

4.23.1.2. BWC Moderate – Initial takeoffs and final landings allowed only when departure and arrival routes avoid bird activity. Local Instrument Flight Rules (IFR)/Visual Flight Rules (VFR) traffic pattern activity is prohibited. Note: When operating at civilian fields without a BASH program, consider the report of "birds in the vicinity" as BWC MOD.

4.23.1.3. BWC Severe – All takeoffs and landings are prohibited. The 55 ECG/CC, or equivalent, may waive this restriction on a case-by-case basis.

4.23.2. When operating at airfields where no BASH program exists, the AC has the authority to delay takeoffs and arrivals due to bird condition after coordinating with the appropriate C2 authority.

4.23.3. Consider bird migratory patterns during the enroute portion of the mission to help minimize the potential of an in-flight bird strike. The Bird Avoidance Model on Air Force Flight Safety Center Flight Safety Division (AFSC/SEF) website contains BASH information including regionalized CONUS bird migration patterns, Portable Flight Planning System software overlay, and the latest news. The Avian Hazard Advisory System website is another source for real time bird hazard information. See AFI 91-212, *Bird/Wildlife Aircraft Strike Hazard (BASH) Management Program*, for additional information.

4.23.4. Following a known bird strike, aircrew should land as soon as practical to have the aircraft inspected. Bird strike damage cannot be accurately assessed in-flight, and undetected damage may result in a complex airborne emergency. AC will contact the nearest safety office within 24 hours for further guidance.

4.24. Traffic Alert and Collision Avoidance System (TCAS). For any resolution advisory (RA) requiring deviation from a clearance, consider filing a Hazardous Air Traffic Report (HATR) upon landing.

4.25. Simulated Emergencies and Simulated Emergency Procedure Training (SEPT). Only conduct simulated emergencies affecting aircraft flight controls and performance in a simulator. Conduct SEPTs in the aircraft with all crewmembers in their seats unless calculated Center of Gravity (CG) permits otherwise.

4.26. Air Shows. The 55 ECG/CC will approve all air shows and establish guidance for static displays.

4.27. Functional Check Flights (FCFs), Acceptance Check Flights (ACFs), & Operational Check Flights (OCFs). FCFs, ACFs and OCFs are accomplished in coordination between 55 ECG/EGV and 645 AESS. If aircrew are asked to accomplish non-standard troubleshooting or checks, contact 55 ECG/EGV.

Chapter 5

AIRCREW PROCEDURES

5.1. Aircrew Uniforms.

5.1.1. Aircrew will wear the aircrew uniform, as outlined in DAFI36-2903, *Dress and Personal Appearance of United States Air Force and United States Space Force Personnel*, and the appropriate MAJCOM supplement on all missions unless otherwise authorized. When the FCG requires civilian attire, dress conservatively.

5.1.2. SQ/CCs will determine clothing and equipment to be worn or carried aboard all flights commensurate with mission, climate and terrain involved. **(T-3)**

5.1.2.1. See AFMAN 11-301, *Aircrew Flight Equipment (AFE)*, for minimum aircrew clothing requirements.

5.1.2.2. Wear of flight gloves for EA-37B aircrew will be at the discretion of the AC.

5.2. Personal Requirements. In addition to AFMAN11-202V3 requirements, aircrew members will carry or wear personal and professional equipment on all flights as follows:

5.2.1. Flight equipment, including as a minimum: headset and operable flashlight.

5.2.2. Identification Tags. Crew members will carry two identification tags on all flights.

5.2.3. A reflective belt or suitable substitute will be worn on flight lines during hours of darkness or periods of reduced visibility.

5.2.4. Driver's License. A valid state driver’s license is required on each Temporary Duty (TDY) where use of United States (U.S.) government general purpose vehicles may be required. Contact the local airfield manager before driving on the flight line.

5.2.5. Shot Record. Crew members will maintain worldwide shot requirements and carry their shot records on all sorties outside the CONUS. This does not apply to combat missions.

5.2.6. Tool Kits. Have at least one tool kit on board for all mission sorties.

5.2.7. Crewmembers will take special care when using and storing extra equipment in the cockpit that is not part of the integral cockpit design. Pilots are to ensure the placement of equipment is secured throughout all phases of flight, night or day, to prevent inadvertent contact with crucial cockpit switches. Furthermore, crew should make all efforts to store additional equipment in the main cabin to prevent inadvertent damage to mission equipment.

5.2.8. Foreign Object Debris Hazards. Crew members will not wear wigs, hairpieces, rings, ornaments, or earrings in the aircraft or on the flight line. Exception: Crew members may wear plain elastic hair fasteners and/or pins, clips, or barrettes providing they do not interfere with the wearing of headsets, or the donning of oxygen equipment. Crew members will account for them before and after flight.

5.3. Pre-Mission Actions. Aircrews will review theater-specific information necessary to successfully operate in the applicable theaters. The review should include (but is not limited to):

5.3.1. Review tasking and itinerary.

5.3.2. Obtain and review applicable OPORD, SPINS, Virtual Risk Assessment (VRA), Country Risk Assessment (CRA) and FLIP.

5.3.3. Review the FCG for areas of operation (to include classified portion). Obtain necessary diplomatic clearances where required.

5.3.4. Obtain required customs forms.

5.3.5. Obtain communications security (COMSEC) materials for the duration of the mission.

5.3.6. Ensure immunizations and all grounding Go/No-Go items are reviewed to ensure they are current and valid throughout the TDY period. CMR items need to be complete IAW ACCMAN11-2EA-37BV1 requirements and Combatant Command (CCMD)/theater guidance such as reporting instructions, SPINS, etc.

5.3.7. Ensure visas have been received, if required.

5.3.8. Obtain terrain charts for unfamiliar destinations if available.

5.3.9. Compile sufficient spare forms, flight orders, etc. to cover the TDY period.

5.3.10. Consider any foreseeable safety risks and risk mitigation factors IAW ORM.

5.4. Aircrew Intelligence Briefing. Before leaving home station on OCONUS missions, aircrews receive an intelligence briefing that emphasizes terrorist, enemy, and friendly political and military development in the area in which they are planned to fly. Obtain timely intelligence updates prior to entering a specific area of operations (AOR). In theater, aircrews should receive intelligence updates on initial arrival at a forward operating location (FOL), or enroute stop, and thereafter when significant developments occur. Report information of possible intelligence value to the local intelligence office as soon as practical to ensure timely dissemination of mission reports (MISREPs).

5.5. Interconnectivity. ACs will ensure interconnectivity with all required planning facilities for the mission. **(T-3)** 55 ECG/HQ is responsible for issuing appropriate communication equipment to ACs for OCONUS operations. Appropriate equipment will be determined based on operational needs and facilitate communication with command and control, maintenance, and planning personnel in case of changes in itinerary. **(T-3)**

5.6. Flight Crew Information File (FCIF).

5.6.1. Crew members will review FCIF, Volume 1, before all missions or ground aircrew duties, and update the FCIF currency record. **(T-3)** Go/No-Go status is IAW AFMAN11-202V2, *Aircrew Standardization and Evaluation Program*, as supplemented. During exercises and contingencies, deployed squadrons will develop procedures to comply with this paragraph and local requirements.

5.6.2. Crew members delinquent in FCIF review or joining a mission enroute will receive an FCIF update from the AC prior to takeoff. **(T-3)**

5.6.3. Crew members not assigned or attached to the unit operating a mission will certify their FCIF review by entering the last FCIF number and their initials behind their name on the file copy of the flight authorization or file copy of their crew orders. This applies to all crew members if the normal sign-in system is not working at show time. IPs flying with general officers or senior staff members are responsible for briefing appropriate FCIF items.

5.7. Aircrew Publications Requirements. Primary crew members will ensure the publications specified in Table 5.1 are carried on all missions. **(T-3)** When the crew includes additional crew members in the same specialty (e.g., more than two pilots on proficiency sorties) each will possess a checklist but otherwise only one set of publications is required. **(T-3)** Units may compile standardized publication kits and issue specific guidance for use to satisfy the requirements of Table 5.1.

Table 5.1. Aircrew Publications.

PUBLICATION	EA-37B
Airplane Flight Manual	P
Airplane Operating Manual (AOM) and Supplements	P
Airplane Performance Handbook	P
Mission Crew Operator Manual (-6)[1]	AMT
Mission Crew Checklist Manual [2]	ALL
Abbreviated Checklists	ALL
AFMAN11-202V3	P
ACCMAN11-2EA-37BV3, *EA-37B Operations Procedures*	P
Appropriate Fuel Planning Documents	P[3]
EA-37B Inflight Guide (IFG)	ALL
Notes: 1. Required on mission sorties only. Stored digitally on mission system disk packs. 2. Stored digitally on the members EFBs as well as mission system disk packs. 3. This is the pilot not identified as AC on the flight authorization.	

5.8. Airfield Review .

5.8.1. Aircrews will consult the web-based airfield database maintained by Air Mobility Command Airfield Sustainability Branch (AMC/A3AS) and comply with the Global Decision Support System 2 (GDSS2)/ Airfield Suitability and Restrictions Report (ASRR) for updates to airfield operability, weight bearing capacity, and Terminal Instrument Procedures (TERPS) reviews. **(T-2)** Refer to AFMAN11- 202V3 and MAJCOM guidance for non-DoW published approach criteria. See MAJCOM/CCMD guidance for foreign instrument procedures not maintained by the U.S. Military, even if they are published in DOW FLIP.

5.8.2. Time and resources permitting, aircrew should verify the currency of respective reports with the web-based GDSS2 application. An acceptable method for accessing airfield Giant Reports sourced from GDSS2 should be the ForeFlight application. If discrepancies arise, aircrew will utilize the most current report and contact respective airfield management representatives for clarification.

5.9. Route Navigation. ACs will ensure navigation planning documents and charts are available during flight IAW Table 5.2. These documents are readily available within the Electronic Flight

Bag within the ForeFlight application. ACs will ensure all pilots on board have up to date FLIP and charts applicable to the mission IAW **Table 5.2**.

5.10. Map/Chart Preparation. Appropriate nav charts which adequately cover planned route of flight should be used on all flights. Each pilot must operate with either electronic (Display Unit or Electronic Flight Bag) or paper charts. Use of electronic maps does not relieve pilots from their responsibility to thoroughly brief all routes, pertinent obstacles, applicable airfield approach control frequencies in the vicinity of class A/B/C/D airspace, intersection of other Visual Routes/Instrument Routes (if applicable), and any other possible areas of conflicts.

Table 5.2. Route Navigation Publications.

Publication
FLIP Planning (GP, AP/1/1B/2/3)
FLIP Instrument Flight Rules (IFR) Supplement
FLIP Flight Information Handbook
FLIP Enroute Charts (High and Low)
FLIP Area Charts (Terminal)
FLIP SIDs/STARs/Approach Procedures
Jeppesen SIDs/STARs/Approach Procedures
Topographical and Sectional Charts
FLIP Visual Flight Rules (VFR) Supplement

5.11. Operations & Mission Products. Carry operations products on all sorties either digitally or physically based on mission requirements. Mission products required for operating the mission system should be defined in the supplement to this volume. Required products include:

5.11.1. Required on all sorties:

5.11.1.1. Computerized flight plan.

5.11.1.2. Flight Authorization (IAW DAFMAN 11-401)

5.11.1.3. AFTO Form 781

5.11.1.4. COMPASS CALL Operational Risk Management Worksheet

5.11.1.5. AF Form 711B, *USAF Mishap Report* (or locally developed Incident Report)

5.11.1.6. DD Form 1801, *DoD International Flight Plan*

5.11.1.7. Weather Information IAW AFMAN 11-202V3, paragraph 4.10

5.11.1.8. Takeoff and Landing Data. A pilot crew member will compute TOLD IAW G550 flight manual as specified **Chapter 10**.

5.11.2. Required on all sorties away from home station, if applicable:

5.11.2.1. Airfield Suitability and Restrictions Report (ASRR)

5.11.2.2. DD Form 1252, *US Customs and Border Protection (CBP) Declaration for Personal Property Shipments*

5.11.2.3. CBP Form 7507, *General Declaration (Outward/Inward) Agriculture, Customs, Immigration, and Public Health*

5.11.2.4. CBP Form 6059B, *Customs Declaration*

5.11.2.5. AF Form 651, *Hazardous Air Traffic Report* (HATR)

5.11.2.6. DD Form 1610, *Request and Authorization for TDY Travel of DoD Personnel*

5.12. Briefing Requirements. Units may amplify these briefing requirements in the supplement to this manual.

5.12.1. Pre-Departure Briefing Items. The AC will contact the local C2 agency to confirm mission requirements. **(T-3)** The AC and controlling agency share joint responsibility to identify special briefing requirements. Briefings may include buffer zone, electronic warfare activities, SAFE PASSAGE, Electromagnetic Interference (EMI), diplomatic clearance, anti-hijacking procedures, operations and safety supplements to flight manuals, and OPORD procedures.

5.12.2. Pilot in Command Briefing. The AC will ensure that an aircrew briefing is conducted prior to engine start for the first sortie of the day. **(T-3)** At a minimum, brief crew members on specific mission details for that day's sortie(s), ORM factors for the mission, and items applicable to aircraft operations including MAJCOM, NAF, and unit special interest items (SIIs). Use briefing guides contained in the 55 ECG In-Flight Guide.

5.12.3. Specialized Briefing. Use specialized briefings to detail operating procedures or SIIs specific to various crew positions, and to answer questions relating to those specialties. Specialized briefings review tactics and procedures, and technical instructions for specialized equipment operations. All crew members should attend each briefing. Crew members may only be excused from specialized briefings for pre-flight duties; however, the AC will back brief all appropriate items. **(T-3)**

5.12.4. Weather Briefings. The AC will obtain a briefing on current weather, trends, and forecast for the proposed route, destination, and alternates. **(T-3)** Verbal briefings are authorized for local flights. The AC will brief primary crew members on appropriate weather conditions before departure. **(T-3)**

5.12.5. Mission Briefing. Conduct mission briefings prior to all mission sorties. Briefing content varies depending on numerous factors including mission requirements, ROE/SPINS, threat assessment, etc. Mission briefings should include, but are not limited, to mission description and purpose, itinerary, aircraft configuration and special equipment, fuel load, clothing requirements, training and evaluation requirements (if applicable), flying safety, and intelligence considerations.

5.12.6. Intelligence Briefings. Before operating in a combat environment, the crew will obtain a current intelligence briefing.

5.12.7. Multi-ship Briefings. In the case of multi-ship operations where an MC is designated, MC-certified aircrew members will lead mass briefings to all crews and ensure that all other required briefings are conducted by their respective crew members.

5.13. Training Aircraft Not Capable of Flight. If an aircraft is not capable of departure within 4 hours after scheduled departure time, the training sortie may launch under the following conditions:

5.13.1. An Operational Risk Management (ORM) analysis will be re-accomplished to consider crew duty day, fatigue, alterations to training objectives and/or profile, aircraft maintenance status, and any other items affecting the mission.

5.13.2. SQ/CC, Operations Officer, or designated representative grants approval.

5.13.3. For planning purposes, a minimum of 1.5 hours should be allotted for aircraft preflight and flight planning duties when determining the anticipated departure time.

5.14. Training/Evaluation Briefings. Before all training/evaluation missions ACs, instructors, and flight examiners will brief their crews on the following additional items:

5.14.1. Training/Evaluation requirements. Instructors and evaluators (for each crew position) will outline requirements and objectives for each student or examinee.

5.14.2. Planned training profiles and seat changes.

5.15. Aircrew Debriefing. Review and evaluate overall training and/or mission performance. Each aircrew member should thoroughly understand the accomplished training and lessons learned from mission employment. Ensure all training is appropriately documented.

5.16. Call Signs.

5.16.1. Training Missions. Use the unit's static call sign prefix followed by a 2-digit suffix assigned by the parent unit.

5.16.2. Operational Missions. Use call signs assigned by OPORD, fragmentation (FRAG), or diplomatic clearance. If call sign is not assigned, obtain and use ACC-assigned off-station call sign. If an ACC-assigned off-station call sign is unavailable, use the Voice Call Sign Listing (VCSL) option. As a last resort, use unit static call signs.

5.17. Flight Plan/Data Verification.

5.17.1. Computer Flight Plan (CFP) Use. CFPs are the official sources of performance, navigation, and climatic data, including enroute wind information. If stand-alone computer-based plans are used, each mission segment should utilize best wind data available. Use only MAJCOM validated CFPs (JMPS, ForeFlight, ARINC).

5.17.1.1. Use CFPs to the maximum extent practical. Flight crews may manually compute flight plans. The AC has final responsibility for flight plan accuracy and diplomatic clearance compliance.

5.17.1.2. Verify CFPs for route of flight and fuel computation accuracy before departure. Pass any flight plan discrepancies.

5.17.2. All waypoint data retrieved from an ***expired*** database should be verified by one or more of the following methods:

5.17.2.1. Latitude/longitude from current FLIP.

5.17.2.2. Bearing/distance from a flight plan after latitude/longitude is verified for each waypoint.

5.17.2.3. Ground Based Navigational Aids (NAVAIDs).

5.17.3. Takeoff and Landing Data (TOLD). A pilot crew member will verify preflight computed TOLD with actual conditions and gross weights. At a minimum:

5.17.3.1. Verify gross weight independently from the TOLD.

5.17.3.2. Verify Flight Management System (FMS) inputs match ambient conditions.

5.17.3.3. Review and compare the computed distances, ground roll, and climb gradient (if applicable) with the actual conditions, runway available, and departure procedures.

5.18. Departure Planning. Use Federal Aviation Administration (FAA) Aeronautical Information Manual (AIM), AFMAN11-202V3, this chapter, and the appropriate MAJCOM supplements. Regardless of the type of departure flown (IFR/VFR), review the following (as appropriate): IFR Departure Procedure, instrument approach plate, NOTAMs, ASRR, and suitable terrain charts.

5.18.1. VFR Departures. Do not fly VFR departures in lieu of obstacle clearance planning. **(T-3)**

5.18.1.1. Conduct an RM analysis for the VFR departure.

5.18.1.2. The minimum climb performance for VFR departures is determined by ensuring all the following conditions are met:

5.18.1.2.1. All-engine climb capability ensures obstacle avoidance along the departure route.

5.18.1.2.2. One-engine Inoperative (OEI) climb capability will ensure departure and emergency return route provides obstacle avoidance.

5.18.1.2.3. In all cases, the aircraft performance must meet or exceed a climb rate of 100 ft/nautical miles (NM) OEI to VFR traffic pattern altitude. **(T-3) Note:** If unable to comply with the conditions stated above, download fuel or delay until more favorable conditions exist.

5.18.1.3. Refer to FLIP for host nation VFR requirements before flying VFR outside of CONUS.

5.18.2. IFR Departures.

5.18.2.1. If the airport does not have an authorized IFR departure method, depart VFR IAW **paragraph 5.18.1**. An IFR departure is not authorized at airfields without an instrument approach.

5.18.2.2. Departure End of Runway (DER) Screen Height. When any doubt exists about which screen height to use, plan to cross the DER at 35 ft (minimum) unless you can ascertain a different screen height requirement from an appropriate authority.

5.18.2.3. IFR Climb-To Altitude. Prior to takeoff, the AC will verify climb-out to a minimum enroute IFR altitude with all engines operating. Furthermore, the AC will ensure that the aircraft can meet all published altitude restrictions for the associated procedure unless ATC has issued a new altitude restriction. The AC will also calculate OEI climb-out performance to an applicable obstacle clearance altitude.

5.18.2.4. Minimum Climb Gradient. In the event the aircraft is unable to meet the published IFR climb gradient with OEI, use the following methods for contingency procedure planning to ensure the aircraft can clear all obstacles along the planned departure route.

5.18.2.4.1. Calculate TOLD using bleeds-off takeoff criteria.

5.18.2.4.2. Ensure current weather meets or exceeds non-standard takeoff minimums to visually clear all obstacles vertically or laterally.

5.18.2.4.3. Reduce fuel load.

5.18.2.4.4. Delay the mission until atmospheric conditions allow for sufficient performance to meet the requirements.

5.18.2.4.5. After exhausting all other contingency procedure planning and the mission is deemed operationally necessary to justify the increased risk, ECG/CC or designated representative may approve the reduction of up to 48 ft/NM from the required IFR climb gradient provided the AC has conducted extensive preflight obstacle planning.

5.18.2.4.6. Depart VFR IAW **paragraph 5.18.1**.

5.18.2.5. Practice Instrument Approaches under VFR. While conducting instrument approach procedures under VFR, crews will comply with the minimum IFR climb gradient for the procedure being flown **(T-3)**

5.18.2.6. Missed Approach Climb Gradients. ACs must ensure the aircraft can meet the published missed approach climb gradient IAW AFMAN11-202V3. **(T-2)** If unable to meet the published minimum climb gradient, crews will comply with **paragraph 5.18.2.4** and subordinate paragraphs.

5.18.3. Critical Field Length (CFL). Takeoff gross weight (GW) considerations must ensure that the runway available is greater than CFL for a normal takeoff. In some cases, a minimum altitude is required at the published screen height. For screen heights greater than 50 ft, increase balanced CFL by 50 ft for every additional foot of required altitude at DER (**Example:** 55 ft screen height drives a 250 ft increase to balanced CFL).

5.18.4. Gross Weight. Unless waived by MAJCOM/A3, ensure that the aircraft does not exceed the maximum GW (91,400 lbs), zero fuel weight (60,000 lbs), or CG limitations (calculated by AC prior to each flight) specified in the aircraft flight manual. GW may be further restricted by operating conditions such as, icing, temperature, pressure altitude, runway length and slope, aerodrome weight bearing capacity, departure maneuvering, required climb gradients, and obstacles.

5.19. Weather Minimums for Takeoff. Minimum Runway Visual Range (RVR) for takeoff is 1600 ft.

5.19.1. Takeoffs are permitted on operational missions with weather greater than or equal to 1000 RVR provided the runway has operating centerline lights, visible runway centerline markings, two operative RVR reporting systems, and both pilots are fully qualified in their respective crew position.

5.19.2. Select a departure alternate when departure weather is below landing minimums IAW **paragraph 5.21**.

5.19.3. If no RVR readout is available for the departure runway, a report of ½ mile (800 meters) or better visibility is required.

5.20. Crosswinds. The maximum crosswind for touch-and-go and full-stop landing is 25 knots.

5.21. Alternate Planning. Select alternate airports IAW AFMAN11-202V3.

5.21.1. Departure Alternates. ACs will select a departure alternate IAW AFMAN11-202V3 and MAJCOM guidance.

5.21.2. Destination Requirements (for filing purposes). The forecasted destination weather meets AFMAN11-202V3 requirements and the following:

5.21.2.1. File two alternates when:

5.21.2.1.1. The forecast visibility (Temporary Group (TEMPO) or prevailing) is less than published for the approved straight-in or sidestep approach.

5.21.2.1.2. The forecast ceiling or visibility (TEMPO or prevailing) is less than published for all other approaches. For approaches with no published ceiling requirement (e.g., Jeppesen approaches), the minimum required ceiling is computed by taking the published height (Height Above Airport [HAA] or Height Above Touchdown/Threshold [HAT/HATh]) and rounding it up to the nearest one hundred feet (or as determined by MAJCOM TERPs review). For example, a Jeppesen VOR approach with a published HAA of 642 ft would require a forecasted ceiling of 700 ft.

5.21.2.1.3. The forecast surface winds (TEMPO or prevailing) exceed RCR-corrected limits.

5.21.2.2. In addition to AFMAN11-202V3 filing requirements for remote or island destinations, the prevailing RCR-corrected surface winds at the destination must remain within demonstrated limits from one hour before to one hour after the ETA.

5.21.2.3. Refer to **Chapter 10** and AFMAN11-202V3 for fuel planning considerations when filing to a remote or island destination, when the destination is located in Alaska or at latitudes greater than 59°N.

5.22. Adverse Weather.

5.22.1. Flight into areas of forecast or reported severe turbulence is prohibited. **(T-3)** Do not fly into an area of known or forecast moderate or greater mountain wave turbulence. **(T-3)**

5.22.1.1. The EA-37B is a Category II aircraft for turbulence. To gain a more accurate picture for their route of flight, crews will confirm the Category type of aircraft to which the forecast turbulence applies to or what Category type of aircraft reported the encounter and convert the turbulence intensity IAW AFH11-203, Volume 2, *Weather for Aircrews—Products and Services*.

5.22.1.2. The PF is responsible for ensuring all personnel on board are seated, with seat belts fastened, when areas of moderate or greater turbulence are encountered or anticipated. **Warning:** ***Serious injury may occur if personnel do not have their seat belts fastened and the aircraft encounters moderate or severe turbulence***.

5.22.2. Flight into areas of forecast or reported severe icing is prohibited. **(T-3)** Prolonged operation, such as cruise flight or holding, in areas of moderate icing should be avoided. Refer

to the AOM for cold weather operations. **Note:** Air Force Weather Agency technical note AFH15-101, *Meteorological Techniques*, states that freezing drizzle is equivalent to moderate icing and freezing rain is equivalent to severe icing.

5.22.2.1. Do not takeoff under conditions of freezing rain. **(T-3)** Do not takeoff under conditions of freezing drizzle except when aircraft has been properly de-iced/anti-iced IAW flight manual procedures. **(T-3)** When freezing fog is forecast or reported, aircrews will confirm with weather agencies what type (if any) icing is associated with the freezing fog.

5.22.2.2. Freezing precipitation, snow, freezing fog, or temperatures near 0°C, may cause ice or frost to accumulate on aircraft surfaces. When an aircraft requires de-icing prior to takeoff, refer to the following:

5.22.2.2.1. Only use de-ice and anti-ice fluids listed in an aircrew's respective flight manual. Aircrews will be familiar with and follow all restrictions in their associated flight manual with respect to anti-ice/de-ice procedures.

5.22.2.2.2. In extreme climatic conditions, anti-ice fluids listed in the AOM are approved for use.

5.22.2.2.3. Most current FAA holdover times for expected duration of anti-ice benefits can be found by initiating a search at the Air Force Flight Standards Agency (AFFSA) website: **https://usaf.dps.mil/sites/affsa**. As a guide for approved anti-icing fluids, crews may use published anti-icing holdover times the AOM.

5.22.2.3. In all cases, ACs will ensure a visual inspection of the aircraft is completed within 5 minutes of departure. **(T-3)**

5.22.3. The AC will exercise sound judgment when a potential to overfly thunderstorms or cumulonimbus clouds exists. Comply with thunderstorm avoidance guidance in AFH 11-203, Volume 1, *Weather for Aircrews*. **Warning:** Regardless of altitude, aircraft damage from thunderstorms is possible at distances in excess of 20 NM. To ensure safety of flight and prevent aircraft damage, aircrews must familiarize themselves with information on thunderstorm development, convective activity, and associated hazards to ensure appropriate avoidance. **(T-2)** Refer to AFH 11-203V1 and FAA AIM, paragraph 7-1-27 for more information.

5.22.4. The use of ground-based radar as a means of thunderstorm avoidance should only be used to assist in departing an inadvertently penetrated area of significant weather. It should never be considered a normal avoidance procedure. When relying exclusively on ground-based radar for weather avoidance, and the ground controller is unable to provide avoidance instructions, attempt to maintain visual meteorological conditions (VMC) by changing routing, diverting to alternate, and/or declaring an emergency and requesting priority assistance.

5.22.5. Aircrews should avoid flying in areas of recently dissipated thunderstorms and clouds due to advection (horizontal movement of clouds caused by wind) downwind of thunderstorms.

5.22.6. To minimize exposure to thunderstorm hazards when approaching or departing an airport in an area where thunderstorms are occurring or are forecast:

5.22.6.1. Attempt to maintain VMC.

5.22.6.2. Maintain at least 5 NM separation from meteorological hazards (e.g. virga, heavy rainfall).

5.22.7. Significant Meteorological Information (SIGMET). National Weather Service in-flight weather advisories are not limiting to AF aircraft. Contact the nearest military weather facility or flight service station (FSS) for details, if applicable.

5.22.8. Volcanic Dust Precautions. Aircraft flight operations in areas of forecast or known volcanic activity or dust is prohibited. Plan all missions to avoid flying downwind of volcanic activity, and in all cases by at least 50 NM.

5.22.9. Lightning Avoidance. The following conditions are most conducive to lightning strikes and prolonged flight in them should be avoided:

5.22.9.1. Within 8°C of freezing level.

5.22.9.2. In clouds or in any intensity of precipitation or turbulence associated with thunderstorm activity.

5.22.10. In clouds within 5,000 ft of the freezing level.

5.23. Flight Progress.

5.23.1. Another pilot or primary crew member verifies waypoint data inserted into the FMS. The PM will verify waypoint names, sequence, courses, altitudes, and distances in the FMS against published FLIP for all flight routings and instrument approach procedures. The PM will notify the crew when the FMS has been properly loaded and verified.

5.23.2. PMs monitoring will use **Chapter 9** procedures for flight following. **(T-3)**

5.24. Communications.

5.24.1. Controller-Pilot Data Link Communications (CPDLC) is the primary means of communication during oceanic operations and should be tested prior to leaving UHF/VHF radio range.

5.24.2. Crews will perform an HF radio ground check before takeoff if HF radio use is required for flight. **(T-3)** Establish HF contact before leaving UHF/VHF radio range.

5.24.3. The aircraft should return/proceed to the nearest suitable support base if CPDLC is inoperative, if unable to establish HF contact with the controlling HF station, and an alternate means of relaying ATC information is unavailable.

5.24.4. When required, pilots will provide ATC position reports and weather observations. If unable to contact an ATC agency, attempt to relay through the global HF stations.

5.25. Descent . Before descent into unfamiliar areas, pilots will review appropriate terrain charts to increase aircrew situational awareness of obstructions. Crew members will only perform their primary duties for aircraft operations, monitor descent and approach, and complete required checklist items from the initial descent point until landing.

5.26. Destination Weather . The AC is responsible for obtaining destination weather, to the max extent practical, prior to the initial descent.

5.26.1. Obtain current weather information from Automatic Terminal Information Service (ATIS) /Automated Surface Observing System (ASOS)/Automated Weather Observation

System (AWOS), any USAF base weather station via Pilot-To-Metro Service (PMSV), Flight Watch, or FSS.

5.26.2. When OCONUS, contact servicing Operational Weather Squadron (OWS) IAW the FIH.

5.26.3. Do not use Air Route Traffic Control Centers (ARTCC) controllers as a primary source for weather information.

5.26.4. **Weather minimums** . When published circling minimums do not include aircraft categories, use a ceiling of 500 ft above the published airport elevation and 1.5 statute mile (SM) visibility or published circling approach minimums, whichever is greater for Category C minimums. For Category D minimums, use a ceiling of 600 ft above the published airport elevation and 2 statute miles visibility or published circling approach minimums, whichever is greater.

5.27. Aircraft Category and Instrument Capabilities.

5.27.1. Approach Category. The EA-37B is a category C aircraft unless higher approach airspeeds are required.

5.27.2. While CAT II Instrument Landing System (ILS) equipment is installed, the EA-37B is not certified for CAT II approaches.

5.27.3. The demonstrated altitude loss for the EA-37B performing a missed approach is 60 feet. Non-precision approaches may be flown in Vertical Navigation (VNAV) provided that a derived decision altitude (DDA) is calculated. (DDA = LNAV MDA(H) + 60 ft).

5.27.4. Area Navigation (RNAV) and Required Navigation Performance (RNP) Operations.

5.27.4.1. EA-37B aircrews are authorized to perform IFR RNAV operations listed in the AFM for which the aircraft is certified and the aircrew is properly trained.

5.27.4.2. The EA-37B RNAV system is suitable as defined in FAA Advisory Circular (AC) 90-108, *Use of Suitable RNAV Systems on Conventional Routes and Procedures*, and may be used as a substitute or alternate means of navigation on conventional routes and procedures IAW AFMAN11-202V3 and G550 AFM guidance.

5.27.4.3. Aircrews are not authorized to fly RNAV (RNP), RNP AR and A-RNP (RF legs) approaches until aircraft equipment is certified, aircrews are trained, and operational approval is issued from ACC A3. This does not affect the capability to fly RNAV, RNAV (GPS), GPS or “OR GPS” approaches.

5.27.5. EA-37B aircraft are certified for RVSM operations. Aircrew will be certified for RVSM airspace operations IAW section 1 prior to operating in RVSM airspace. Comply with the following requirements when operating in RVSM airspace:

5.27.5.1. Both primary altimeters, at least one autopilot, the altitude advisory system, and the transponder, must be fully operational.

5.27.5.2. Engage the autopilot during level cruise, except when circumstances such as the need to re-trim the aircraft or turbulence require disengagement.

5.27.5.3. Periodically cross-check the primary altimeters to ensure they agree within 200 feet. When operating in surveillance airspace (Radar/Automatic Dependent Surveillance-

Broadcast), the initial altimeter cross-check should be performed after level-off. During oceanic and remote continental operations, a cross-check should be performed and recorded in the vicinity of the point where oceanic and remote continental navigation begins (e.g., on coast out).

5.27.5.4. Limit climb and descent rates to 1,000 feet per minute when operating near other aircraft to reduce potential TCAS advisories.

5.27.5.5. If unable to comply with RVSM requirements (aircraft equipment failure, weather, lost com, etc.), the crew must notify ATC and request a new clearance as soon as the situation allows.

5.27.5.6. Cold Weather Altimeter Corrections. Crews will reference the AFMAN11-202V3 and G550 AFM when applicable.

5.28. In-Flight Emergency (IFE) Procedures .

5.28.1. Inform ATC, C2 agencies and MEPs of an IFE and crew intentions when time and conditions permit.

5.28.2. The AC may initiate a CONFERENCE HOTEL when additional expertise is necessary. Reference the In-Flight Guide for current in-flight contact information.

5.29. Cockpit Voice Recorder (CVR) . If involved in a mishap or incident, after landing and terminating the emergency, pull the CVR ("VOICE RCDR" circuit breaker in the Aircraft Operating Manual) and Flight Data Recorder (FDR) power circuit breakers.

5.30. Dropped Objects . If an externally dropped object is discovered, the flight crew will:

5.30.1. Notify ATC or the controlling agency as soon as practical and include details of routing, altitude, weather, etc.

5.30.2. Notify maintenance at the first transited military station.

5.31. Impoundment of Aircraft. If an aircraft is involved in a serious in-flight incident, the AC will immediately impound the aircraft after landing and contact the controlling C2 agency for further instructions.

5.32. Risk Management (RM) . RM is a logic based, common sense approach to making calculated decisions on human, material, and environmental factors before, during, and after all operations. ACs will accomplish RM worksheets IAW MAJCOM and local guidance as part of preflight activities. Deployed crews will comply with local instructions and use locally derived RM worksheets. In the absence of deployed guidance, the 55 ECG RM worksheet may be used.

5.33. Maintenance.

5.33.1. Complete the Aircraft Forms after each flight. After landing, crew members will debrief maintenance personnel, if available, on aircraft condition, engines, avionics equipment, and all installed special equipment as required.

5.33.2. Document anytime the aircraft is flown under 1000 ft above ground level (AGL) over saltwater except for takeoffs and landings. Place an entry in the Aircraft Forms stating, "Aircraft Subjected to Salt Spray" (state lowest altitude and duration).

5.34. Aircraft Forms.

5.34.1. Review Aircraft Forms before applying power to the aircraft or operating aircraft systems. Sign for aircraft acceptance before flight. Ensure that the DD Form 1896*, DoD Fuel Identaplate*, is aboard the aircraft and Aviation Into-Plane Reimbursement Card (AIR Card) information available for servicing. The aircraft must have a physical AIR Card on board. **(T-3)**

5.34.2. One-Time Flights. An aircraft may be released for a one-time flight with a condition that might be hazardous for continued use, provided the aircraft is airworthy for one flight to a repair facility. Refer to T.O. 00-20-1, *Aerospace Equipment Maintenance Inspection, Documentation, Policies, and Procedures*, for downgrade authority and procedures. After the maintenance release is obtained, coordinate mission requirements with the controlling agency. The AC's concurrence is required before the aircraft can be flown.

5.34.3. Refer to **Chapter 10** for NMC clearing procedures at stations without maintenance support.

5.35. Aircraft Servicing and Ground Operations.

5.35.1. Aircraft Refueling. Aircrew members certified in ground refueling or specialized refueling operations may perform refueling duties. Certified crew members may augment maintenance refueling teams at enroute stops.

5.35.2. Preflight Inspection Requirements.

5.35.2.1. The aircraft dash one preflight inspection remains valid until either:

5.35.2.1.1. Aircraft ground time exceeds 12 hours (72 hours provided the aircraft is sealed, not flown and documented entry control is maintained).

5.35.2.1.2. Another maintenance dash six preflight is performed.

5.35.2.2. When an aircrew assumes a pre-flighted spare or quick turn, a thorough visual inspection will be performed. **(T-3)** A thorough visual inspection includes, but is not limited to, ensuring all panels are secure, tires and struts are inflated, all hydraulic reservoirs are serviced, and there are no visible fluid leaks on the aircraft.

5.35.2.2.1. An aircraft should be sealed if the aircrew preflight has been completed and the aircraft is left unattended by aircrew. The AC will maintain positive control of the aircraft keys once an aircraft sealing operation is complete.

5.35.2.2.2. Once the aircraft is sealed, the following notes are entered in the Aircraft Forms. Notify CLS what time the aircraft is sealed and seal number.

5.35.2.2.2.1. **Note:** FAOM preflight inspection C.W., Time:, Fuel:, GOX:, Seal #:

5.35.2.2.2.2. Write seal numbers in ink. State what sections of the preflight were not accomplished.

5.35.2.2.3. Only a maintenance officer, maintenance superintendent, production superintendent, pilot or AMT may break an aircraft seal. When the seal is broken, annotate the reason for breaking the seal in the Aircraft Forms. Complete the date corrected block and sign the corrected/transferred by block. Additionally, a new discrepancy block is completed as follows:

5.35.2.2.4. Aircraft Resealed, Seal #, See Page: , Block: (this refers to the discrepancy block where the aircraft was originally sealed).

5.35.3. Fire Protection and Crash Rescue.

5.35.3.1. The aircraft engine fire extinguisher system fulfills the minimum requirements for fire protection during engine start.

5.35.3.2. At AC discretion, a fireguard may be utilized for engine starts including the Auxiliary Power Unit (APU). A crew member or ground support personnel may act as fireguard.

5.35.4. Towing. If towing operations are conducted by personnel unfamiliar with EA-37B or Gulfstream G550 towing procedures, the AC will coordinate with the senior maintenance representative or superintendent to ensure the towing supervisor and crew are qualified. **(T-3)** At non- USAF installations, the AC must have approval from the airfield operations officer or airfield manager prior to towing. **(T-3)** The AC will ensure the tow team supervisor briefs all personnel on their duties and the associated hazards. **(T-3)** Proper checklists will be used. If any doubt exists as to the qualification of tow team personnel or the safety of the operation, make no attempt to tow the aircraft until qualified personnel can be located.

5.35.5. Exercise caution when entering G550 Operating Manual defined “Danger and Caution Areas.”

5.36. Aircraft Recovery Away from Main Operating Base. The AC is responsible for ensuring the aircraft is turned to meet subsequent mission tasking. If qualified aircraft maintenance specialists are unavailable, the aircrew is responsible for turning the aircraft to meet subsequent mission taskings.

5.36.1. The AC is responsible for the recovery items including:

5.36.1.1. Parking and receiving.

5.36.1.2. Aircraft servicing, including Aircraft Ground Equipment (AGE) usage.

5.36.1.3. Supervision of minor maintenance within local capability.

5.36.1.4. Minor configuration changes to meet mission tasking.

5.36.1.5. Securing the aircraft before entering crew rest.

5.36.1.6. Coordinating aircraft security requirements.

5.36.1.7. Documenting Aircraft Forms.

5.36.2. In all cases when aircrews will service the aircraft without qualified maintenance specialist assistance, comply with the appropriate maintenance T.O.

5.36.3. When maintenance personnel are unavailable, the aircrew will document the overdue maintenance inspection (e.g., preflight, basic post-flight, calendar inspection) in the Aircraft Forms. If the current aircraft status is not a red dash symbol, enter a red dash symbol in the Aircraft Forms.

5.37. Aircrew Flight Equipment (AFE) Requirements.

5.37.1. Oxygen. Prior to takeoff, sufficient on-board oxygen will accomplish the planned flight from the equal time point (ETP) with a minimum of 1500psi in the system and all walk-around bottles filled.

5.37.2. Aircrew members will comply with AFMAN11-202V3 oxygen requirements. Additional crew members not required for mission accomplishment may be considered passengers when determining oxygen requirements.

5.37.2.1. The EA-37B is equipped with an Automatic Emergency Descent Mode (when Autopilot is engaged above FL400), available and approved pilot quick-don style masks which are automatically adjusted upon donning.

5.37.2.2. Pilot Oxygen Regulators must be set to 100% when aircraft is operated above FL410.

5.37.3. Crew members occupying a crew station will have an oxygen mask with communication connected and readily available prior to engine start until engine shutdown. **(T-3)**

5.37.4. Life preserver units (LPUs) or Personal Floatation Devices. Each crewmember will ensure an LPU and anti-exposure suit, when required by AFMAN 11-301, is within easy reach before takeoff on overwater flights. **(T-3)**

5.37.5. AFE Documentation. When Survival Backpack or other AFE equipment (JSAM, Night Vision Device (NVD), Aircrew Laser Eye Protection (ALEPs), etc.) are present, ACs will ensure all prepositioned AFE and survival equipment items are serviceable, inventoried, and certified on the AFTO Form 46, *Prepositioned Aircrew Flight Equipment*. Notify AFE of any onboard equipment shortages or unserviceable conditions. Annotate discrepancies in the Aircraft Forms. ACs will ensure AFE equipment inspections will remain current at enroute stops when a crew change is not anticipated. Do not open sealed bags to inventory equipment. Inspect each sealed bag's serviceability tag for equipment list and inspection currency.

5.38. Border Clearance.

5.38.1. Normal Operations.

5.38.1.1. The unit dispatching the mission is normally responsible for the border clearance of its aircraft.

5.38.1.2. The AC shares responsibility for border clearance. When an EA-37B aircraft is on-loaded at a base without an air traffic function, the AC is responsible for ensuring the following:

5.38.1.2.1. Crew members possess current passports and valid visas, as required.

5.38.1.2.2. Crew members have current immunization certificates (shot records).

5.38.1.2.3. Departing or entering the U.S. through a Port of Entry where border clearance can be obtained.

5.38.1.2.4. Obtaining border clearance for and baggage, as required, before takeoff to a foreign area or after arrival from a foreign area.

5.38.1.2.5. Spraying the aircraft (see the FCG and **paragraph 5.36**.).

5.38.2. Procedures for U.S. Entry.

5.38.2.1. Enroute, the AC will distribute personal customs declarations (CBP Form 6059B) to all crew members, brief crew members on customs regulations, and prepare and compile necessary border clearance forms. **(T-3)**

5.38.2.2. Enroute, notify the C2 agency at the base of intended landing of any change in ETA to ensure that border clearance is accomplished as soon as possible after landing.

5.38.2.3. Obtain a permit to proceed, when military necessity requires the aircraft which has landed in the U.S. for customs clearance, to proceed to another U.S. base to obtain border clearance. A permit to proceed delays customs inspection of cargo and crew until arrival at the offload station, and saves intermediate offloading and reloading normally required for customs inspection. A permit to proceed is valid only to the airport of next landing where the border clearance will be completed unless a new permit to proceed is issued by a customs official. Do not make intermediate stops between a permit to proceed issuing point and the manifested cargo destination unless required due to an emergency or when directed by the C2 controlling agency.

5.38.2.4. When an aircraft lands for a U.S. border clearance, a U.S. Customs representative will normally meet the aircraft to obtain the required documents. Do not deplane crew members unless necessary for safety or the preservation of life and property. **Exception:** Pilots may deplane to perform necessary duties for ground operations. Do not unload until approved by customs and agriculture personnel or their designated representatives. This procedure applies to the initial landing in the U.S. and all landings required when operating on a permit to proceed or until all crew and cargo complete final border clearance.

5.38.3. Inspections of U.S. Aircraft by Foreign Officials.

5.38.3.1. Follow USAF policy on status of military aircraft as stated in DoW FCG. In summary, this policy holds that U.S. military aircraft are immune from searches, seizures, and inspections (including customs and safety inspections) by foreign officials. Additionally, ACs should be aware of, and adhere to, any specific FCG provisions for individual countries.

5.38.3.2. If confronted with a search request by foreign authorities, aircrews should use the following procedures.

5.38.3.2.1. In most cases, search attempts may be halted simply by the AC stating to the foreign official that the aircraft is a sovereign instrumentality not subject to search without consent of USAF headquarters or the U.S. Department of State officials in the country concerned. This should be clearly conveyed in a polite manner so as not to offend foreign authorities that may honestly, but mistakenly, believe they have authority to search USAF aircraft.

5.38.3.2.2. If foreign authorities insist on conducting a search, the AC should make every effort to delay the search until he or she can contact USAF headquarters (through MAJCOM C2) or the appropriate embassy officials. The AC should then notify these agencies of foreign request by the most expeditious means available and follow their instructions.

5.38.3.2.3. If foreign officials refuse to desist in their search request, pending notification to USAF headquarters or the appropriate embassy, the AC should indicate that he/she would prefer to fly the aircraft elsewhere and request permission to do so (provided fuel, flying time, and mechanical considerations permit a safe flight).

5.38.3.2.4. If the request is denied and the foreign authorities insist on forcing their way aboard the aircraft, the AC should state that he/she protests the course of action being pursued and that he/she intends to notify both USAF headquarters and the appropriate American embassy of the foreign action. The AC should not attempt physical resistance and should thereafter report the incident to USAF headquarters and appropriate embassy as soon as possible. The AC should escort foreign authorities if the inspection cannot be avoided.

5.38.3.3. Other procedures may apply when carrying sensitive cargo or equipment. Follow these procedures and applicable portions of classified FCG supplements.

5.38.4. Exercises and Contingency Operations.

5.38.4.1. General. Certain missions, which do not transit normal Ports of Entry, require special procedures to expedite compliance with customs, public health, immunization, and agricultural requirements. A joint memorandum of understanding between these agencies and MAJCOM establishes certain procedures and waivers.

5.38.4.2. Implementation. Implementation of the agreement is not automatic. Traffic and border clearing agencies implement all or part of the agreement as necessary for each operation. Inspection and clearance may be accomplished at the U.S. on-load or off-load base, or at the foreign on-load or off-load base.

5.38.4.3. Customs Procedures.

5.38.4.3.1. Outbound: No requirement. Filing of CBP Form 7507 is not required unless directed.

5.38.4.3.2. Inbound: Prepare one copy of the following documents before arrival: CBP Form 7507 (Passenger list not required).

5.38.4.4. Public Health Procedures.

5.38.4.4.1. When operating from a base without a traffic officer, the AC ensures all crewmembers are properly immunized.

5.38.4.4.2. Spray the aircraft if required.

5.38.4.5. Immigration Procedures.

5.38.4.5.1. Outbound: No requirements.

5.38.4.5.2. Inbound: Submit the following to the immigration inspector if carrying civilian passengers.

5.38.4.5.3. One copy of CBP Form 7507.

5.38.4.6. Agriculture Procedures:

5.38.4.6.1. Outbound: No requirement.

5.38.4.6.2. Inbound: Consult Border Clearance Guide.

5.38.5. Military Customs Pre-clearance Inspection Program. All crew members will follow Military Customs Pre-clearance requirements. Expect a Customs representative will meet the aircraft and collect all declarations.

5.39. Communications Instructions for Reporting Vital Intelligence Sightings (CIRVIS) and Other Reports. Vital intelligence sightings in flight will be reported IAW FLIP planning or FLIP Enroute Supplement. Report other incidents as indicated in AFMAN 10-206, *Operational Reporting (OPREP).*

5.40. Insect and Pest Control.

5.40.1. Responsibility. ACs will ensure required spraying is accomplished IAW Department of War FCG, or as directed by higher headquarters. Certify the spraying on CBP Form 7507, or on forms provided by the country transited. The only exception is when mandated by the FCG.

5.40.1.1. When spraying is required, use insecticide, aerosol d-phenothrin-2 percent, National Stock Number (NSN) 6840-01-067-6674 (or equivalent), to spray the aircraft. Wear leather or Nomex gloves while spraying.

5.40.1.1.1. Direct the nozzle toward the ceiling of the compartment or space being sprayed.

5.40.1.1.2. Spray spaces inaccessible from within the aircraft after completely loading fuel, baggage, cargo, including baggage compartments, wheel wells, and other similar spaces.

5.40.1.1.3. Spray the cabin, flight deck, and other spaces accessible from within the aircraft after the crew is aboard and after closing all doors, windows, hatches, and ventilation openings. **Caution:** If the insecticide label directs disembarkation after use, spray before boarding crew. Close all doors and hatches for 10 minutes after dispensing and ventilate for 15 minutes before allowing anyone on board.

5.40.1.2. Spray for 105 seconds unless longer periods are specified for the country being transited. **Note:** Keep used aerosol cans separate from other trash so they may be disposed of safely.

5.40.2. AC Responsibility In-flight. When seeing any insect or rodent infestation of the aircraft in-flight, notify the destination C2 center, airfield management operations, or airport manager of the situation before landing so the proper authorities can meet the aircraft.

5.40.3. Procedure at Aerial Port of Debarkation (APOD). On arrival at an APOD, do not open cargo doors or hatches except to enplane officials required to inspect the aircraft for insect or rodent infestation.

5.41. Aircrew Flight and Dash 21 Equipment Documentation. The AC, or designated representative, is responsible for:

5.41.1. Before departing home station and enroute stations, ensure appropriate serviceable protective clothing, AFE, survival, and Dash 21 equipment are aboard the aircraft for the entire or remainder of the mission.

5.41.2. Before departing home station and following enroute crew changes, review AF Form 4076, *Aircraft Dash 21 Equipment Inventory*, to ensure maintenance has certified the

installation of all required Dash 21 equipment, maintenance has signed the initial inspection, and configuration documents match mission requirements.

5.41.3. Before departing home station and following enroute crew changes, review, sign, and date the AFTO Form 46 to ensure AFE has certified the installation of all required protective clothing, AFE and survival equipment, and that configuration documents match mission requirements. Ensure appropriate number and type of life preservers are aboard for over-water missions.

5.41.4. Missing Equipment. Aircrew members discovering missing equipment will accomplish the following:

5.41.4.1. Make an Aircraft Forms entry for the missing equipment. Additionally, ensure equipment removed from the aircraft at an enroute station is documented in the Aircraft Forms.

5.41.4.2. Annotate AF Form 4076 and AFTO Form 46 in the next vacant column to indicate the quantity remaining for the item. Enter the International Civil Aviation Organization (ICAO) location designator above the column's check number. Leave AF Form 4076 and AFTO Form 46 aboard the aircraft in the event of an enroute crew change.

5.41.4.3. Advise the AC and determine whether the missing equipment should be recovered or replaced before mission continuation.

5.41.4.4. Assist in preparing reports of survey for missing equipment, as required.

5.41.4.5. Advise CLS, operations supervisor (or equivalent) and appropriate C2 agency (or airport management) before mission continuation.

5.41.5. Additional Equipment. If undocumented additional equipment is discovered during the preflight, annotate the total quantity on the AF Form 4076 or AFTO Form 46 in the next vacant column for the item. Ensure the ICAO location designator is entered above the column's check number.

5.42. Classified Equipment and Material.

5.42.1. Equipment. When classified equipment is onboard, ensure the C2 center or airfield management operations office is aware of the requirement for aircraft security IAW **Chapter 6**. At bases without USAF jurisdiction, ensure the aircraft and equipment are protected. DoDM 5200.01V1_DAFMAN16-1404V1*, Information Security Program: Overview, Classification, and Declassification*, provides specific guidance concerning the security of classified equipment aboard aircraft. For classified aircraft components which cannot be removed and stored, lock and seal the aircraft. If available, use security forces to guard the aircraft; otherwise, use guards employed by the host country for flight line/airport area control. Zeroize navigation and radio equipment if left unattended.

5.42.1.1. DAFI 31-101, *Base Defense Operations*, and **Chapter 6** define COMPASS CALL aircraft security requirements. Security requirements should also be specified in the deployment order (DEPORD) for deployed operations.

5.42.1.2. For additional equipment installed onboard for operation or training missions, aircrews will be familiar with and adhere to specific device/equipment handling,

processing and storage procedures. Aircrew should contact equipment provider and 755 OSS/OSK for guidance prior to usage.

5.42.2. Material. Ensure COMSEC and other classified materials are turned in at destination and receipts are obtained for COMSEC and classified material. The on-site C2 center may provide temporary storage for COMSEC and other classified materials during enroute, turnaround, and crew rest stops. Prior to depositing classified material for over-night storage, the AC must ensure the facility is authorized for classified storage equal to the material's classification level and on-site personnel are qualified to handle the classified material being deposited. Encrypted COMSEC will only be transferred to authorized DoD personnel. The AC and MCC will ensure that all material, discussions, and display screens are limited to the clearance level of escorted personnel.

5.42.3. Mode 5 Procedures.

5.42.3.1. Aircrews will ensure Mode 5 is operable when required for mission accomplishment. Aircrews will conduct a Mode 5 operational ground test (ground test assets permitting) before overseas deployment or as specified in the OPORD and/or contingency/exercise tasking.

5.42.3.2. Attempt to fix an inoperable Mode 5 before takeoff. Do not delay takeoff or cancel a mission for an inoperable Mode 5 except when planning to transit an area where safe passage procedures are implemented.

5.42.3.3. Conduct a Mode 5 operational check in-flight on all missions departing the CONUS. Aircrews can request a Mode 5 interrogation check with North American Aerospace Defense Command (NORAD) on UHF frequency 364.2.

5.42.3.4. Aircraft with inoperable Mode 5 may continue to their intended destination. Repairs will be accomplished at the first destination where equipment, parts, and maintenance technicians are available. In theaters where safe passage is implemented, aircraft will follow procedures for inoperable Mode 5 IAW the applicable airspace control order or Air Tasking Order (ATO) and/or SPINS.

5.42.3.5. Mode 5 operational checks on the ground or in-flight are a mandatory maintenance debrief item. The AC will annotate successful and unsuccessful interrogations of the Mode 5 in the Aircraft Forms.

5.42.4. When required for mission accomplishment, aircrews will carry COMSEC equipment and documents for Mode 5 operation IAW SPINS. Crews will contact their local COMSEC manager for guidance before departure for destinations without COMSEC storage facilities. **(T-3)**

5.42.5. Emergency Destruction. If possible, destroy/damage classified material and/or equipment prior to a crash landing.

5.42.6. Ground Egress. If ground egress is required and classified material cannot be secured, the AC or designated representative will obtain the names and telephone numbers of all un-cleared personnel who boarded the aircraft. If classified material was inadvertently disclosed, contact the Special Securities Officer (SSO).

Chapter 6

AIRCRAFT SECURITY

6.1. General.

6.1.1. This chapter provides guidance for aircraft security and unlawful seizure of aircraft.

6.1.2. Protection Levels. EA-37B aircraft are Protection Level (PL) 3 assets but become Protection Level 2 when 270VDC power is applied to aircraft mission systems and mission system drives are installed. The aircraft remains PL 2 for 60 minutes following de-powering of the aircraft mission system. See DAFI 31-101 for definitions and designations of Protection Levels.

6.2. Security. See AFI13-207-O, *Preventing and Resisting Aircraft Piracy (Hijacking)(FOUO)*, AFI 10-701, *Operations Security (OPSEC)*, and DAFI 31-101, for requirements for protection of aircraft in transient status at U.S. and foreign bases.

6.3. Security Procedures.

6.3.1. Forward Operating Location (FOL) Security. DETCOs or Advanced Echelon (ADVON) personnel will make security arrangements at FOLs and will comply with the minimum requirements IAW DAFI 31-101.

6.3.2. Enroute Security. Aircrews will receive a threat assessment and security capability evaluation briefing at home station and receive updates at enroute C2s. If necessary, assess the situation and perform the following actions:

6.3.2.1. Request area patrol coverage from local security forces. If local authorities request payment for this service, consult appropriate C2 channels.

6.3.2.2. Direct armed aircrew members to remain with the aircraft and maintain surveillance over aircraft entrances and activities in the vicinity of the aircraft. Acquire a means to report suspicious or hostile activity to security forces, if available.

6.3.2.3. If the AC determines airfield security is inadequate and the safety of the aircraft is in question, (e.g., local security forces is not sufficient or unavailable and the crew is not augmented with security forces) the AC may waive crew duty time limitations and depart as soon as possible for a base where adequate security is available. If departure is not possible, the aircrew must secure the aircraft to the best of their ability. Crew rest requirements are subordinate to aircraft security when the airframe/equipment may be at risk. Request security assistance from the nearest DoD installation, U.S. embassy, local military, or law enforcement, as appropriate.

6.3.2.4. If the AC determines the aircraft must be locked and sealed to detect unauthorized entry, use the aircraft lock, and secure the hatches, windows, and doors in a manner that will identify unauthorized entry. Prior to departing the aircraft, wipe off the immediate area surrounding the lock and latches to aid an investigation if a forced entry is discovered upon return. Coordinate procedures with local base operations aircraft servicing while the crew is away. Report any unauthorized entry or tampering to the Office of Special Investigation, security forces and/or local authorities, and the C2 agency. Thoroughly inspect the aircraft prior to flight.

6.4. Arming of Aircrew Members. Due to the nature of the EA-37B mission, crews are not normally armed enroute to a FOL. However, weapons may be carried on the aircraft for use in theater. If the ECG/CC determines the nature of the deployment warrants the aircrew to carry weapons enroute, the following procedures will be utilized.

6.4.1. Weapons Issue (Enroute). When required by ROE, ATO, or SPINS, obtain weapons and ammunition from the weapons storage area. Crew members will be armed IAW AFI31-117, *Arming and Use of Force by Air Force Personnel*, and MAJCOM directives. Present a current DAF Form 522, *Ground Weapons Training Data*, for weapon issue. The same weapon is reissued until the mission terminates. If an armed aircrew member leaves the crew enroute, transfer the weapon to another authorized aircrew member using DAF Form 1297, *Temporary Issue Receipt*.

6.4.1.1. Load and unload weapons at approved clearing barrels if available. To transfer loaded weapons to another aircrew member, place the weapon on a flat surface. Do not use hand-to-hand transfer.

6.4.1.2. Do not wear weapons off the flight line except to and from the armory and other facilities associated with aircrew activities (e.g., base operations, fleet service, cargo and passenger terminals, flight line cafeteria or snack bar). Crew members will maintain positive control of the weapon at all times.

6.4.1.3. Arm aircrew members prior to preflight duties.

6.4.1.4. Store weapons in the most secure facility available prior to crew rest (e.g., base or civil law enforcement armory). If a weapons storage facility is unavailable, secure firearms and ammunition in the aircraft. If the aircraft is not equipped with a gun box, leave the weapons in the most secure and least visible location on the aircraft. Attempt to seal the weapons with a tamper-evident seal (e.g., boxcar seal) and maintain the seal number. Lock and seal the aircraft doors when leaving the aircraft.

6.4.2. Contingency Missions.

6.4.2.1. Crewmembers will be issued weapons with their survival equipment IAW theater directives prior to combat/combat support sorties.

6.4.2.2. The SQ/CC or DETCO, in conjunction with AFE personnel, will establish procedures for weapons issue. Additionally, deployed leadership will develop in-garrison weapons issue/arming procedures commensurate with current force protection procedures. **(T-3)**

6.5. Preventing and Resisting Hijacking. Refer to AFI13-207-O for detailed guidance. Security operations at deployed locations are normally sufficient to deter aircraft piracy. Aircrew should always remain vigilant and report any unusual activity to security forces.

6.6. Armed Passengers. EA-37B aircraft do not normally carry passengers. Therefore, the risk of hijacking is further reduced. While onboard, passengers do not normally possess weapons or ammunition. Exceptions include special agents and guards of the Secret Service or State Department or other individuals specifically authorized to carry weapons. Take every precaution necessary to prevent an accidental weapons discharge.

6.6.1. Passengers or deadhead crewmembers will not retain custody of ammunition on an aircraft and will turn it in to the troop commander or AC. **(T-3)** Passengers authorized to carry

weapons may carry unloaded weapons and ammunition aboard the aircraft during combat operations.

6.6.2. If guards or couriers need to clear their weapons, the AC will ensure the individual:

6.6.2.1. Moves to a safe, clear area at least 50 ft from any aircraft, equipment, or personnel before un-holstering or unslinging their weapons.

6.6.2.2. Clears weapons IAW standard safety procedures.

6.7. Force Protection. Crews will always remain alert for possible terrorist activities. Reference AFTTP3-4, *Airman's Manual* and Department of Defense Instruction (DoDI) O-2000.16V1_DAFI31-145-O, *Antiterrorism (AT) Program Implementation*, for force protection measures.

Chapter 7

OPERATIONAL REPORTS AND FORMS

7.1. General. This chapter contains a description of applicable reports and forms. For assistance in completing safety forms contact the wing, unit, or local flight safety officer.

7.2. DAF Form 457,*Hazard Report***(DAFI 91-202,***The Department of the Air Force (DAF) Mishap Prevention Program***).** AF hazard reporting system provides a means for Air Force personnel to alert supervisors and commanders to hazardous conditions requiring prompt corrective action. A hazard is any condition, act, or circumstance that jeopardizes or may jeopardize the health and well-being of personnel, or which may result in loss, damage, or destruction of any weapons system, equipment, facility, or material resource.

7.3. AF Form 651, *Hazardous Air Traffic Report* (HATR).

7.3.1. The Air Force HATR program provides a means for personnel to report all near midair collisions (NMAC), TCAS resolution advisories requiring the aircraft to deviate from assigned course/altitude, alleged hazardous air traffic conditions, or NAVAIDS, FLIP or published directions/instructions that contributed to a hazardous situation. Use information in HATR reports only for mishap prevention.

7.3.2. DAFI 91-202 and DAFMAN 91-223, *Aviation Safety Investigations and Reports*, list HATR reportable incidents and procedures.

7.4. AF Form 711B,*USAF Mishap Report Worksheet***(DAFI 91-204*Safety Investigations and Reports*).**

7.4.1. Responsibilities. Notify the appropriate authorities of any mishap involving aircraft or crew. When notified, appropriate authorities initiate investigative and reporting actions IAW DAFI 91-204. **Note:** Do not attempt to classify a mishap.

7.4.2. Reportable Mishaps.

7.4.2.1. Report damage to the aircraft, or injury to the crew; also report any damage or injury to another organization’s equipment or personnel resulting from the movements or actions of an aircraft or crew.

7.4.2.2. Report the following occurrences:

7.4.2.2.1. A physiological episode is a physiological reaction, near accident, or hazard in-flight due to medical or physiological reasons. This includes:

7.4.2.2.1.1. Aircrew decompression sickness from evolved gas (the bends, chokes, skin, neurological, or neurocirculatory manifestations).

7.4.2.2.1.2. Aircrew loss of consciousness or incapacitation in-flight.

7.4.2.2.1.3. Aircrew hypoxic (altitude) hypoxia (suspected, probable, or definite).

7.4.2.2.1.4. Aircrew trapped gas disorders (ear, sinus, teeth, or abdominal).

7.4.2.2.1.5. Aircrew symptoms or health effects caused by toxic, noxious, or irritating materials such as smoke, fumes (including carbon monoxide) or liquids.

7.4.2.2.1.6. All events during which a crewmember executed any portion of an

emergency checklist in response to toxic smoke/fumes/liquid exposure.

7.4.2.2.1.7. When the aircraft unintentionally strikes another object regardless of damage cost or need for repairs.

7.4.2.2.1.8. Loss of cabin pressurization that requires executing an emergency checklist.

7.4.2.2.1.9. Aircrew G-induced loss of consciousness.

7.4.2.2.1.10. Aircrew spatial disorientation of any type (including visual illusion) resulting in an unusual aircraft attitude.

7.4.2.2.1.11. Any medical condition, event or physical injury directly resulting from performance of flight activities that an aeromedical professional determines is significant to the health of the aircrew.

7.4.2.2.1.12. Suspected Laser Exposure. If exposed to a laser, the AC notifies appropriate C2, intelligence, safety and medical agencies as soon as possible. Aircrew who suspect exposure to laser radiation from either friendly or hostile sources should report to the FS's office or nearest emergency room where individual can be examined by an ophthalmologist immediately upon landing. Reference AFI 48-139, *Laser and Optical Radiation Protection Program*.

7.4.2.2.1.13. Hyperventilation.

7.4.2.2.1.14. Death by natural causes of any aircrew member during flight.

7.4.2.2.1.15. Unintentional loss of pressurization if cabin altitude is above FL180, regardless of effects on personnel.

7.4.2.2.1.16. Alcohol and hangover (crew only).

7.4.2.2.1.17. Illness (both acute and pre-existing), including food poisoning, dehydration, myocardial infarction, seizure, and so forth. **Note:** In the event of a physiological episode, all involved aircrew members will report to a FS as soon as practical and request that an AF Form 711B be accomplished.

7.4.2.2.2. Emergency, precautionary, or inadvertent engine shutdown at any time after taxi until normal engine shutdown. Report any loss of thrust sufficient to prevent maintaining level flight at a safe altitude. **Note:** FCFs are excluded; however, report failure to restart, using the criteria above.

7.4.2.2.3. Flight control malfunction resulting in an unexpected or hazardous change of flight attitude, altitude, or heading.

7.4.2.2.4. All un-commanded flight control inputs whether it results in a dangerous situation or not. Report autopilot and/or autothrottle faults if, in the opinion of the investigator, the fault would have put the aircraft in a dangerous situation.

7.4.2.2.5. Structural failure of critical landing gear components. Malfunction of landing gear when difficulty is experienced using emergency system or procedures.

7.4.2.2.6. In-flight loss of all pitot-static instrument indications or both primary and standby attitude indicators.

7.4.2.2.7. In-flight fires, massive fuel leakage in an engine bay, all gear-up landings.

7.4.2.2.8. Spillage or leakage of radioactive, toxic, corrosive, or flammable material from aircraft stores that creates a hazardous condition or an airborne emergency divert.

7.4.2.2.9. All cases of departure from intended takeoff or landing surface onto adjacent surfaces.

7.4.2.2.10. Any incident which does not meet the established criteria for a reportable mishap but, in the judgment of the AC, needs to be emphasized in the interest of flight safety.

7.5. Petroleum, Oil and Lubricants (POL)—Aviation Fuels Documentation. This section prescribes aviation POL (AVPOL) procedures that ensure correct documentation, form and invoice processing, and program supervision. Reference AFMAN11-202V3. Use the Multi Service Corporation (MSC) AIR Card for the purchase of aviation fuel and ancillary ground services at commercial airports (and some military installations) worldwide. The AIR Card is authorized for use by all U.S. government, state, local law enforcement, and some foreign government aircraft. All ACs should plan to use the AIR Card. In most cases, there should not be fees when refueling at non-Defense Logistics Agency Energy (DLA Energy) contract locations. The MSC card is accepted at approximately 4,800 locations worldwide. It replaces the Standard Form (SF) 44, *Purchase Order-Invoice Voucher (Storage Safeguard Form)*, at locations that accept the MSC card.

7.5.1. Responsibilities. Aircrew and maintenance personnel must be familiar with AVPOL procedures and documentation requirements of this chapter. Purchasers may be financially liable for improper use of the AIR Card.

7.5.2. Refuel/de-fuel USAF aircraft at DOW locations to the max extent possible. If DOW service is not available, purchase fuel from other source(s) in the following priority:

7.5.2.1. DLA Energy or Canadian into-plane contracts.

7.5.2.2. Foreign government air forces. **Note:** Historically, the most reliable source of contract servicers has been **https://aircardsys.com** or alternatively **https://www.airseacard.com**.

7.5.2.3. Open market AIR Card purchases to include SHELL™ International Trading Company (SITCO) agreement.

7.5.3. DD Form 1896, *DoD Fuel Identaplate*. The DD Form 1896 will be utilized at military installations to purchase fuel unless automated data capture (ADC) equipment is installed. The AC or designated representative will complete the form, log the purchase and place a copy inside the AF Form 664, *Aircraft Fuels/Ground Servicing Documentation Log*.

7.5.4. Refueling at Locations other than USAF Bases.

7.5.4.1. DD Form 1898, *Energy Sales Slip*. This form is used to record the aviation fuels transaction (issue or defuel) at other DoD locations, including into-plane contract locations. Log and place the DD1898 inside the AF Form 644. The AC or designated representative will complete this form. **Note:** If the contractor insists on a unique invoice along with the DD 1898, annotate the vendor's invoice with "DUPLICATE DD1898 ACCOMPLISHED".

7.5.4.2. AF Form 664 is a tool to log and store all AVPOL transaction forms. Record all off station transactions on the front of the form and insert the original form inside the envelope. Turn in the AF Form 664, with supporting forms, to maintenance debriefing or as directed by local procedures. The AC or designate representative will complete this form when appropriate.

7.5.4.3. Purchasing Aviation Fuel in Canada. The DoD and Canadian Department of National Defense have signed a memorandum of understanding allowing DoD aircraft to use the DD Form 1896 when refueling at Canadian airfields with a Canadian National Defense Contract (CNDC). Use the AIR Card for fuel purchases at Canadian airports without a CNDC, and for ground handling services at all Canadian airports.

7.5.4.4. Use host country forms to effect purchases at foreign military airfields, including replacement-in-kind locations. Hand scribe information from DD Form 1896 on the local form. Log and place a copy inside the AF Form 664.

7.5.4.5. SF44, *U.S. Government Purchase Order-Invoice-Voucher*. This may be used to purchase fuel, ground services and/or other authorized products when no MSC card contract is in place.

7.5.4.5.1. SF44 fuel purchases where Fixed Base Operations (FBO) agrees to invoice DLA Energy for payment.

7.5.4.5.1.1. The aircrew presents the SF44 as the purchase invoice when an FBO refuses to accept the MSC card. The aircrew completes the SF44 and attaches it to the FBO vendor ticket/invoice when the FBO also declines use of the SF44 and uses its own invoice/receipt. Fuel purchases are documented on a separate SF44 from ground services and other authorized products since the FBO invoices DLA Energy for the fuel and the customer for non-fuel product and services.

7.5.4.5.1.2. Copies 1 and 2 of the SF44 are provided to the FBO. Copy 1 of the SF44 and one copy of the FBO commercial invoice, if applicable, is forwarded to the following address by the FBO to bill/invoice DLA Energy: DLA Finance Energy F8-FR, 1525 Wurtsmith Street, Building 5730, JBSA-Lackland, TX 78236 or email: **DESCSALocalPurchase@dla.mil**.

7.5.4.5.1.3. Copy 3 of the SF44 and one copy of the FBO commercial invoice, if applicable, is provided to the aircrew. Log and place a copy inside the AF Form 664. Aircrews present all fuel purchase receipts to the designated aviation squadron Certifying Official and/or Accountable Official upon return to home station to enable timely validation and financial obligation processing into the Fuels Automated System (FAS). **Note:** AIR Card. The AIR Card is a commercial credit card which allows aircrews to purchase aviation fuel, fuel related supplies, and/or ground services at commercial airports where no DOW/Canadian into-plane contracts exist. Accepted at over 4200 locations, it is intended to replace the SF44 at locations that accept the AIR Card. All AF aircraft are issued an AIR Card.

7.5.4.5.2. SF44 fuel purchases where the FBO requires cash payment.

7.5.4.5.2.1. Cash fuel purchases are only authorized when either the Electronic Foreign Clearance Guide requires cash payment or when FBO locations outside the

U.S. and U. S. Territories refuse the MSC card and/or SF44 invoicing process. Aircrews will employ the following procedures when required to pay cash for aviation fuel purchases. **Note:** These procedures do not apply to non-fuel products or services.

7.5.4.5.2.2. . The aircrew will obtain cash from a local DOW Finance source that will be charged to an approved Treasury suspense account prior to home station departure.

7.5.4.5.2.3. Aircrews complete the SF44 and obtain the FBO fuel vendor annotation in block 11 of the form to confirm total cash amount. Sign and date blocks 20 and 21. Log and place a copy inside the AF Form 664. Aircrew will return unused cash to their local DOW Finance source upon return to home station.

7.5.4.5.3. SF44 purchases of ground services and other approved products (not fuel). Complete a separate SF44 for non-fuel purchases. Provide the FBO copies 1 and 2 of the SF44. The FBO uses copy 1 and one copy of the FBO commercial invoice, if applicable, to directly bill/invoice the purchasing organization. Block 9 of the SF44 reflects the organization name and address of the finance office responsible for payment to the FBO. The purchasing organization makes payment to the FBO upon receipt of the invoice from the FBO. Log and place a copy inside the AF Form 664.

7.5.4.5.4. If the vendor presents their own form for signature and accepts the SF44, write the statement “SF44 Executed” on the vendor’s form.

7.5.4.5.5. Turn in two copies of the SF44 to the operations officer at home station.

7.5.4.5.6. Present the DD Form 1896 for purchases at SITCO agreement locations. Make certain the invoice includes date of transaction, grade of product, quantity issued/defueled, unit of measure, and signature of AF member who accepted product. If vendor also requires completed SF44 write statement, “AF FORMS EXECUTED on vendor’s invoice”. Log and place a copy inside the AF Form 664.

7.6. Aircraft Forms. Use Aircraft Forms to record POL actions for specific airframe IAW applicable directives. The AC or designated representative completes the form and turns it in to maintenance debrief.

Chapter 8

AIRCREW OPERATIONS IN CHEMICAL BIOLOGICAL, RADIOLOGICAL, NUCLEAR, AND HIGH-YIELD EXPLOSIVE THREAT ENVIRONMENT (CBRNE)

8.1. Overview. The proliferation of Chemical, Biological, Radiological, Nuclear and High-Yield Explosive (CBRNE) weapons and the means to deliver them present serious security threats to all airborne operations. See AFMAN 11-301 for general aircrew operational guidance and DAFI 10-2503, *Chemical, Biological, Radiological, or Nuclear (CBRN) Defense Program*, for general CBRNE discussion.

8.2. CBRNE Passive Defense Measures. Passive defense measures are those activities conducted to negate, contain, and manage the effects of CBRNE attack. Passive defense measures include pre-, trans-, and post-attack actions designed to mitigate the CBRNE threat through contamination avoidance, protection, and contamination control.

8.2.1. In-flight Divert. When advised that a destination airfield is under CBRNE attack or has been contaminated, the aircrew will divert to an uncontaminated airfield, if possible. Authority to land at a contaminated airfield is specified in the controlling OPORD.

8.2.2. Survival Launch. If on the ground during an attack warning, make every reasonable effort to survive provided the aircraft has sufficient fuel and unrestricted/safe access to the runway. In practice, this option may only be practical for aircraft that have just landed or aircraft at or near the end of the runway. If launch is not possible, shut down engines and avoid running environmental control systems. Close aircraft doors/hatches, don IPE and seek personal protective cover on the base. For missile attacks when time does not permit using base facilities, remain in the sealed aircraft for a minimum of one-hour after the attack and/or follow host base guidance.

8.2.3. Individual Protective Equipment. The current in-flight protective gear issue for aircrew members is listed in AFMAN 11-301.

8.3. Ground Operations.

8.3.1. Establishing the Threat Level. Aircrews should monitor C2 channels to ensure they receive the latest information concerning the destination's alert condition. Diversion of aircraft to alternate "clean" locations may be required, unless operational necessity otherwise dictates. The theater C2 agency (normally through the controlling OPORD) directs aircrew pre-exposure activities.

8.3.2. Crew Rest Procedures. Operational necessity may require the aircrew to rest/fly in a contaminated environment. If the mission is not being staged by another aircrew or pre-flight crews are not available, the aircrew may pre-flight, load, and secure the aircraft prior to entering crew rest. The departing aircrew will perform necessary crew preparations and pre-flight briefings. If possible, aircrew transport should be provided in a covered vehicle. To prevent spreading contamination to personnel and/or the aircraft, aircrews should avoid pre-flighting the aircraft prior to departure. As aircrews proceed to fly, they will require assistance from ground support personnel in removing their aircrew protective over cape and over boots prior to entering the aircraft.

8.3.3. Mission Planning. Mental preparation is key for aircrews facing the dangers of CBRN weapons. Thorough flight/mission planning is required. ACs should emphasize (Aircrew Chemical Defense Ensemble (ACDE) wear, crew coordination, CBRNE hazards and countermeasures, in-flight divert, plans for on- load/off-load in the event of a ground attack, and plans for the return leg in the event of aircraft contamination. Alternative scenario plans should also be considered in the event mission- oriented protective posture (MOPP) conditions change.

8.3.4. Oxygen Requirements. Operating a contaminated aircraft may increase oxygen requirements. ETM-EC-37B-AOM contains charts of oxygen consumption if operating at cabin altitudes above 10,000 ft MSL.

8.3.5. Donning Equipment.

8.3.5.1. ACDE. Aircrew don ACDE based on the alarm condition (See AFTTP 3-4). Use the “buddy dressing” procedures and refer to AFE- provided donning checklist to ensure proper wear. When wearing the ACDE, Atropine and 2 Pyridine Aldoxime Methyl (PAM) Chloride auto injectors will be kept in the upper left ACDE pocket. If the integrated survival vest/body armor is worn, the Atropine and 2 PAM Chloride auto injectors may be kept in the lower right flight suit pocket. This standardized location enables personnel to locate the medication should an individual be overcome by chemical warfare agent poisoning.

8.3.5.2. M-9 Paper. M-9 paper on the flight suit facilitates detection of liquid chemical agents and Aircrew Contamination Control Area processing. M-9 paper should be placed on the flight suit prior to entering the CBRN threat area or when an Alarm Yellow or higher has been declared.

8.3.5.3. AC Responsibilities and brief when inbound to a CBRN. When inbound to a CBRN threat area, prior to descent, the AC will ensure crew don appropriate protective equipment IAW arrival destination’s MOPP level and brief aircrew operations in the CBRN threat area. As a minimum, include the following in this briefing: flight deck isolation, oxygen requirements, air conditioning system requirements, IPE requirements, ground operations and MOPP levels. Aircrew members determine if the wear of the integrated survival vest/body armor and LPUs restrict dexterity and mobility to the point that it becomes a safety issue. If the aircrew deems the equipment to create a safety of flight concern, then the items may be pre-positioned (instead of worn) on the aircraft to be readily available to the aircrew.

8.3.6. On-load and Off-load Considerations. Exercise extreme care to prevent spreading contamination to the aircraft interior during ground operations, particularly to the cockpit. Reduce the number of personnel entering the aircraft. Do not place contaminated engine covers, safety pins and chocks in the aircraft unless sealed in clean plastic bags and properly marked IAW T.O. requirements. Aircrew members entering the aircraft will remove plastic over boots and over cape portions of the aircrew ensemble and ensure flight/mobility bags are free of contaminants and placed in clean plastic bags. Prior to entering the aircraft all personnel should implement boot wash/decontamination procedures. Aircrew exiting aircraft into a contaminated environment will don plastic over boots and over cape prior to leaving the aircraft.

8.3.7. Communications. Conducting on-/off-loading operations, while wearing the complete ACDE, complicates communications capability. Use the mini-amplifier/speaker or the aircraft public address system and augment with flashlight and hand signals, as required.

8.3.8. 10-Foot Rule. The 10-foot rule was developed to provide guidance for protecting personnel using or handling contaminated resources (such as pallets) or working in locations with materials that might retain a residual chemical. The 10-foot rule embodies a safety factor that goes beyond current Office of the Secretary of War (OSW) guidance (which allows removal of IPE whenever detectors no longer detect a chemical agent vapor hazard). There are two phases associated with the 10-foot rule.

8.3.8.1. Initial Phase. During the initial phase, personnel remain in MOPP 4 whenever they stay within 10ft of the contaminated equipment for more than a few seconds. This MOPP level provides personnel the maximum protection from the chemical agent as it transitions from a contact and vapor hazard to a vapor hazard only.

8.3.8.2. Follow-on Phase. In the follow-on phase, personnel use gloves (e.g., leather, rubber, cloth) when operating on or handling the contaminated equipment. Although a contact hazard is unlikely, relatively small amounts of the agent may still be present. The use of gloves ensures that unnecessary bare skin contact with agent residue is avoided.

8.3.8.3. **Table 8.1** shows "estimated" times associated with initial and follow-on phases of the 10-foot rule. To simplify response processes, commanders may choose to use the worst case scenario as the foundation for all 10-foot rule actions, e.g., 24 hours for the initial phase and all periods of time greater than 24 hours for the follow-on phase.

8.3.8.4. Additional Threats. Blood agents damage mask filters. All personnel will change mask filters at the earliest possible opportunity after a blood agent attack. **Exception:** Filters installed in aircrew CWU-80/P filter packs can only be removed and replaced by AFE personnel.

Table 8.1. "10-Foot Rule" Time Standards.

Agent	Initial Phase	Follow-on Phase
HD	0-12 hrs	Greater than 12 hrs
GB	0-12 hrs	Greater than 12 hrs
GD, GF, GA	0-18 hrs	Greater than 12 hrs
VX, R33	0-24 hrs	Greater than 24 hrs
Note: Rule is based on expected contamination of an airbase following a chemical attack. Adjust times if agent concentration is higher than expected.		

8.4. Flight Operations.

8.4.1. Outbound with Actual/Suspected Chemical Contamination. Air washing is a useful in-flight decontamination technique for removing most of the liquid agent from aircraft metal surfaces. However, vapor hazards may remain in areas where the airflow characteristics prevent complete off-gassing (e.g., wheel wells, flap wells, rivet and screw heads, joints).

Flights of at least 2 hours are recommended, and lower altitudes are more effective than higher altitudes. Fly with the aircraft configured (gear and flaps extended) as long as possible to maximize the airflow in and around as many places as possible.

8.4.2. Once airborne with actual/suspected vapor contamination, purge the aircraft for 2 hours using Smoke and Fume Elimination procedures. To ensure no liquid contamination exists, conduct a close inspection of aircrew, passenger ensembles, and cargo using M-8 and M-9 detection paper. Detection paper only detects certain liquid agents and does not detect vapor hazards. Above the shoulder ACDE should only be removed if there is absolutely no vapor hazard. Be advised that residual contamination (below the detectable levels of currently fielded detection equipment) may be harmful in an enclosed space. The aircrew must take every precaution to prevent spreading of liquid contaminants, especially on the flight deck area. The best course is to identify actual/suspected contamination, avoid those areas for the remainder of the flight, and keep the cargo compartments cool. If an aircrew member has been in contact with liquid contaminants, all personnel aboard the aircraft will stay in full ACDE/ground crew ensemble (GCE) until processed through their respective contamination control area. Upon arrival, park the contaminated aircraft in an isolated area and cordoned to protect unsuspecting ground personnel.

8.4.3. Documenting Aircraft Contamination.

8.4.3.1. When it is suspected or known that an aerospace vehicle or piece of equipment has been contaminated with a radiological, biological or chemical contaminant, an NMC condition is entered and an annotation is made in historical records for the lifecycle of the equipment. Before clearance of an NMC condition for contamination, consult Bioenvironmental Engineer or higher DoD authority.

8.4.3.2. Limits of Decontamination. Complete decontamination of aircraft and equipment may be difficult, if not impossible, to achieve. Crews must restrict formerly contaminated assets to DoD-controlled airfields and not released from US government control. **(T-3)**

8.4.4. Communicating Down-line Support. Pass aircraft contamination information through command and control channels when inbound. This information is used to determine if a diversion flight is required or decontamination teams are needed. Report the physical condition of any crew who are showing agent symptoms and whether they are wearing chemical defense ensembles.

Chapter 9

NAVIGATION PROCEDURES

9.1. General. This chapter contains EA-37B navigation procedures and forms. It is used in conjunction with procedures and requirements set forth in AFMAN11-202V3 and FLIP. Since airspace and associated navigational equipment capabilities are rapidly evolving, aircrew need to maintain an in-depth knowledge of current policies and requirements.

9.2. Mission Planning.

9.2.1. Flight Planning Navigation Database. Flight plans created using commercial flight planning tools (ARINCDirect^{SM}) use their own commercial navigation databases. Pilots are responsible for ensuring the accuracy of flight plan waypoints against current aeronautical charts, terminal instrument procedures, or FLIP documents.

9.2.2. Predictive Receiver Autonomous Integrity Monitoring (RAIM). The FMS uses predictive RAIM to determine the integrity levels at specific locations/times to support a non-precision approach and pilot's flight planning activities. The Global Navigation Satellite Systems (GNSSUs) have the following types of RAIM predictions:

9.2.2.1. Destination.

9.2.2.2. Alternate waypoint.

9.2.2.3. Approach area.

9.2.3. The destination and alternate waypoint predictions are made at specific locations or they are the estimated time of arrival (ETA) when the FMS makes the request for flight planning purposes. Satellites can be manually deselected or enabled for the destination and alternate waypoint predictions. The approach area RAIM prediction is an output of current RAIM projected 5 minutes into the future. Approach area RAIM is a continuous output that is performed without any interaction with the FMS.

9.3. Capability/Airspace Requirements.

9.3.1. Reduced Vertical Separation Minimum (RVSM). Airspace (FL290-FL410) where RVSM is applied is considered special qualification airspace. The EA-37B is fully equipped and approved for operations in these areas. **(T-2)** Commanders will ensure pilots are appropriately trained and approved for operation within RVSM airspace. Refer to FLIP Area Planning documents for theater unique information and the AFMAN11-202V3, paragraph 4.28 for RVSM requirements.

9.3.2. Performance Based Navigation (PBN). Both the operator and the specific aircraft type require approval for operations in defined airspace. PBN airspace is being incorporated around the world to increase air traffic capacity by decreasing separation requirements between routes. EA aircraft are approved for en route RNP, but limited to operational time restrictions based on navigation equipment. These limitations are annotated in local supplement and updated by unit Stan/Eval when appropriate.

9.3.3. Frequency Modulation Immunity (FMI). EA aircraft are equipped with dual FMI VHF navigation receivers, considered fully compliant, and follow normal procedures. Refer to the applicable Area Planning series for more information concerning FMI operations.

9.3.4. CPDLC Operations. Refer to applicable FLIP, FAA, Global Operational Datalink (GOLD) Manual, and MDS guidance for CPDLC operations. Specific operations and procedures are approved by MAJCOM/A3 with mission execution authority. Verify equipment and ensure appropriate code is annotated on the flight plan IAW FLIP General Planning (GP).

Chapter 10

FUEL PLANNING AND CONSERVATION

10.1. General. This chapter is designed to assist pilots and planners in fuel planning for missions. A fuel plan is required for all flights except local area training flights with established standard fuel loads. The flight plan and aircraft T.O./Performance Manuals are the primary preflight references. The planning procedures in this chapter apply to all MAJCOM-directed and local missions. Missions should be planned at altitudes, routes, and airspeeds to minimize fuel usage.

10.2. Fuel Conservation. It is Air Force policy to conserve aviation fuel when it does not adversely affect training, flight safety or operational readiness. Aircrew and mission planners manage aviation fuel as a limited commodity and precious resource. Consider fuel optimization throughout all phases of mission planning and execution. Comply with the following whenever consistent with T.O. guidance and safety.

10.2.1. Fuel Loads. Excessive ramp and recovery fuel adds to aircraft gross weight and increases fuel consumption. Do not ferry extra fuel beyond optimum requirements for safe mission accomplishment and training objectives.

10.2.2. Flight Planning. Aircrew and mission planners optimize flight plans and flight routing for fuel efficiency.

10.2.3. APU Usage. Minimize the APU usage to the maximum extent possible. Early coordination may be required to ensure external power carts and heating/cooling units are available.

10.2.4. Center of Gravity (CG). Load and maintain aircraft at an aft CG whenever possible consistent with mission requirements and Flight Manual restrictions.

10.2.5. Engine Start. Delay engine start on all departures whenever practical to minimize fuel consumption.

10.2.6. Taxi. An engine out taxi may be considered when permitted by the Flight Manual. However, due to aircrew unfamiliarity with this procedure it should only be used when situation dictates.

10.2.7. Departure Planning. Consider use of opposite direction runway to reduce taxi and/or expedite departure routing if winds allow.

10.2.8. Takeoff. Don’t delay gear and flap retraction.

10.2.9. Climb/Descent. In-flight procedures such as climb/descent profiles and power settings should also be considered for efficient fuel usage.

10.2.10. Weather Deviations. Attempt to coordinate for off-course deviation early so gross maneuvering is not required.

10.2.11. Cruise techniques. Attempt to trim the aircraft and match throttle settings whenever possible. Normal cruise speeds for the EA-37B is 0.78-0.80 Mach. Use Long Range Cruise (LRC) when required for mission accomplishment. Use high speed cruise only when needed to satisfy mission requirements. Do not exceed Federal Aviation Regulation (FAR) or host nation aircraft speed restrictions.

10.2.12. Approach. Fly most direct routing to arrival approach consistent with mission requirements. Delay initial configuration as much as practical after considering approach complexity, weather, pilot proficiency, etc.

10.2.13. Holding. If holding is required, hold clean at the most fuel efficient altitude and request a large holding pattern. Hold at endurance or performance manual recommended holding speeds, conditions permitting.

10.2.14. Parking. Consider using shortest taxi route and avoid double blocking when able.

10.3. Fuel Planning Procedures. Aircrew should employ the following aviation fuel optimization measures without compromising flight safety or jeopardizing mission/training accomplishment:

10.3.1. Plan fuel to an alternate only when AFMAN 11-202V3 or this manual requires the filing of an alternate.

10.3.1.1. When only one alternate is required, use the closest suitable airfield meeting mission requirements and AFMAN 11-202V3 weather criteria.

10.3.1.2. If two alternates are necessary, use the two closest suitable airfields meeting AFMAN 11-202V3 weather criteria. They should not be within the same terminal area, if terminal forecasts are marginal. Plan fuel to the more distant of the two.

10.3.2. Reserve and contingency fuel is calculated IAW AFMAN 11-202V3, as supplemented.

10.3.3. Fuel Tankering. Tankering fuel for convenience is prohibited. Tankering fuel may be required due to tight turn times and limited fuel availability. When transiting certain high fuel cost locations, there is a significant opportunity to avoid excessive fuel charges by tankering fuel. Consider tankering fuel if it does not negatively impact mission accomplishment.

10.4. Fuel Requirements. This section augments AFMAN 11-202V3 fuel requirements.

10.4.1. Required Ramp Fuel Load (RRFL). As a minimum, required ramp fuel consists of all fuel required for engine start, taxi, warm-up, APU operation, takeoff, climb, cruise, en route reserves (if required), alternate/approach/missed approach (if required), descent, approach and landing.

10.4.2. Aircraft Specific Fuel Planning. Use the following guidelines for developing a RRFL:

10.4.2.1. APU, Start, Taxi, Takeoff Fuel: 500 lbs.

10.4.2.2. Approach and Landing Fuel: 500 lbs.

10.4.2.3. Normal recovery fuel is the fuel on initial or at the final approach fix at the base of intended landing or alternate, if required.

10.4.2.4. Minimum fuel required upon landing at destination or landing at alternate (if an alternate is required IAW AFMAN 11-202V3): 3,000 lbs. This includes the required reserve and contingency fuel IAW **paragraph 10.3.3**.

10.4.2.5. Joker fuel is a pre-briefed fuel state needed to terminate an event and transition to the next phase of the sortie.

10.4.2.6. Bingo fuel is a pre-briefed fuel state that allows the aircraft to return to the base of intended landing or an alternate, if required, using preplanned recovery parameters and arriving with normal recovery fuel.

10.5. Equal Time Point (ETP). First Suitable Airfield (FSAF) and Last Suitable Airfield (LSAF) are used in the ETP calculation. These are represented as the First Nearest and the Last Nearest airports in the ETP calculation. They are airports closest to the coast out and coast in waypoints that meet applicable destination alternate requirements except weather. Forecast weather conditions for LSAF/FSAF (ETA from the ETP/DF +/- 1 hour) will meet or exceed minimums for the lowest compatible approach or 500/1, whichever is greater **(T-2)**. CFPs should plan to arrive overhead at the recovery location with 0+30 minutes reserve fuel. Additional fuel (e.g., descent, approach and landing) may be required, based on mission requirements.

10.6. Planning Factors for Local Flights. When planning fuel requirements for local flights, consider planned flight training times, planned ground times between locals and planned minimum landing fuel requirements. Local training flights may be scheduled for more or less flying time with the proper coordination. IPs and Evaluator Pilots (EPs) will initiate an approach to a full stop landing when fuel on board is equal to or less than the amount specified as the minimum fuel for landing **(T-2)**. Request priority handling from air traffic control if necessary.

10.7. Required Ramp Fuel for EROPS Capable Aircraft. Block to block fuel must be greater than or equal the fuel required to fly to the ETP, experience a loss of cabin pressure, and proceed from the ETP to a recovery field at 10,000 feet or 13,000 feet (if sufficient crew oxygen is available) **(T-2)**. If extra fuel is necessary, it is added and identified extra. **Note:** Flight may be planned and flown using recovery from the ETP at Single Engine Service Ceiling (up to FL 250), instead of 10,000 feet.

Chapter 11

COMBAT MISSION PLANNING

11.1. General. Planners and aircrews should reference their respective AFTTP 3-3 volume for additional mission planning guidance. Refer to the following AFTTP 3-1 for details pertaining to combat mission planning:

11.1.1. AFTTP 3-1.IPE, Tactical Employment IPE (S), is maintained online at 561st Weapons Squadron (561 WPS) website.

11.1.2. AFTTP 3-1.Threat Guide, Individual Chapters (S), are maintained online at 547th Intelligence Squadron website.

11.1.3. AFTTP 3-1.EC-130H/EA-37B, Tactical Employment (S), is maintained online at 561st Joint Tactics Squadron website.

11.2. Responsibilities.

11.2.1. Both the AC and MCC share responsibility for mission planning. When an MC is designated, the MC is responsible for mission planning on missions involving multiple EA-37B aircraft.

11.2.2. Deployment Planning Team (DPT). When a DEPORD is received, the 755 Operational Support Squadron (755 OSS) will establish a DPT to support squadron deployments. The DPT plans the aircraft movement to the deployed location and subsequent employment until an MPC is established. ACC Air Operations Squadron assistance may be utilized if applicable.

11.2.3. MPC. The MPC conducts mission planning for COMPASS CALL employment. Wartime MPC concepts provide an integrated team of aircrew, intelligence, and computer support personnel. The deployed commander will activate the MPC and assign a team chief. The deployed operations officer determines MPC augmentation requirements. The team chief will assign roles and responsibilities to individual team members IAW operational needs. MCs should lead the MPC to the max extent possible.

11.2.4. Intelligence Support. Intelligence briefings are presented IAW ACCMAN 14-402, *Unit-Level Intelligence Mission and Responsibilities*, as supplemented, and intelligence operating instructions.

11.2.4.1. Upon activation, intelligence personnel will support the DPT with an initial situation/threat briefing and provide update briefings as necessary. **(T-3)** They will also provide an intelligence pre-deployment briefing to aircrews. **(T-3)**

11.2.4.2. Intelligence personnel will support the MPC by providing an initial situation/threat briefing, extracting pertinent ATO information, analyzing threats to mission aircraft, building/maintaining a target database and establishing targeting priorities IAW mission directives. They will provide continuous, in-depth analysis of the situation and update the MPC. Intelligence personnel will conduct the intelligence portions of mission brief and debrief.

11.2.4.3. Intelligence personnel will serve as a liaison between COMPASS CALL, the Intelligence, Surveillance and Reconnaissance (ISR) community and mission package planners. **(T-3)**

11.2.4.4. Intelligence personnel will provide COMPASS CALL Mission Crew Simulator (CCMCS) support for scenario development, scenario briefing and scenario debriefing. **(T-3)**

Chapter 12

TACTICAL/THREAT AVOIDANCE PROCEDURES

12.1. General. Use these procedures as well as the flight manual when operating into airfields where a known or suspected ground threat exists. In a threat environment, aircrew members will understand the limitations of themselves and their equipment. The following procedures are not all encompassing. Therefore, aircrews should use good judgment and sound airmanship to successfully accomplish the mission.

12.1.1. This chapter deals primarily with the takeoff/departure and approach/landing phase of flight. AFTTP 3-1.EC-130H/EA-37B (S) contains a more detailed discussion of threat avoidance for all phases of flight.

12.1.2. Carefully consider performance data and energy management, particularly in mountainous terrain at heavy gross weights or with less than full engine capability. Failure to manage energy levels may cause a stall or require a go-around. Consideration should be given to planning increased airspeeds. Another accepted technique is to calculate and post the stall speeds for 0, 30 and 60 degrees of bank as well as the single engine service ceiling. **Warning:** ***Uncoordinated flight reduces stall margins and can cause an abrupt departure from controlled flight***. **Caution:** Uncoordinated flight increases airframe structural loading and should be avoided unless an actual threat exists.

12.1.3. Use all available navigational aids, within threat and emission control requirements, to remain oriented to aircraft position. Aircrew members share equal responsibility for enroute navigation, terrain avoidance and threat lookout. Attention should be focused outside the aircraft, emphasizing threat detection and situational awareness. Limit duties which may cause distractions from lookout procedures to mission essential (ME) items only.

12.2. Approaches. Plan approaches IAW AFTTP 3-1.EC-130H/EA-37B (S) and the airfield identification procedures published in the OPORD or SPINS. Where multiple options are available, select the approach which best minimizes exposure to the threat while still allowing a high probability of landing on the first approach. Remain unpredictable. Plan on intercepting the glidepath no later than three-quarters of a mile prior to the touchdown zone. Limit bank angles to 30 degrees below 1,000-feet AGL at night and below 400-feet AGL during daytime.

12.2.1. Low Altitude Approaches. Use these approaches primarily when a low altitude ingress is necessary, (e.g., radar surface to air missiles en route to the field). All maneuvering is done at low altitudes. Enter these approaches from any direction at en route altitude and airspeed.

12.2.2. High Altitude Approaches. Use these approaches when a high or medium altitude ingress is necessary, (e.g., small arms environment) and allow some reconnaissance of the field via fly over. Base initial altitude, airspeed and heading on the threat.

12.3. EnginesRunning Onload/Offload (ERO) Procedures. Preparation and a thorough briefing enhance the ability to operate quickly and safely. Brief appropriate ground personnel and subsequent aircrews on unexpected hazards encountered during takeoff or landing, e.g., dust, winds, hostile activity. Accomplish the ERO IAW applicable Flight Control Manual (FCM) and AFTTPs.

12.4. Combat Entry and Exit Checklists. As detailed in the MDS-specific AFTTP 3-1, Combat Entry and Exit Checklists should be checklist inserts for use on training and operational missions into simulated or actual threat environments. Expanded Combat Entry and Exit Checklists with detailed explanations/instructions should be included in the Aircrew Tactics Guide for in-flight reference. In the absence of specific directives, the Combat Entry Checklist is initiated prior to entering the threat environment. The Combat Entry Point is determined by theater guidance or known/suspected weapons engagement zone (WEZ). In the absence of more specific directives, initiate this checklist no later than 30 minutes prior to entering the threat environment. If the aircraft departs an airfield within a threat environment or enters the threat environment within 1 hour after takeoff, the checklist should be run prior to the line-up checklist. The Combat Exit Point is determined by theater OPORD/SPINS guidance or known/suspected WEZ.

12.5. COMPASS CALL Hand-Off Guide. This checklist is a locally derived aid intended to streamline coordination between two COMPASS CALL aircrews when one aircraft is replacing the other aircraft on a designated orbit.

Chapter 13

SEARCH AND RESCUE

13.1. General. The EA-37B crew will defer to their C2 agency for prioritization between their tasked mission and the Combat Search and Rescue (CSAR) mission. **(T-2)** However, the crew will be prepared to act as on-scene commander until CSAR forces arrive. Crews then assist CSAR forces in any way necessary. The following general instructions apply to all search missions:

13.1.1. Brief the mission purpose to crew members unable to attend the operations briefing.

13.1.2. Reference AFTTP 3-1.EC-130H/EA-37B (S) for additional guidance.

13.2. Crew Duties. EA-37B crews will allocate crew duties IAW their pre-mission planning. If re-tasked to a CSAR mission, crews should allocate crew duties as follows:

13.2.1. MCC Responsibilities:

13.2.1.1. Plan the search with the mission crew.

13.2.1.2. Discuss radio communication procedures with the crew.

13.2.1.3. Communicate mission de-confliction with C2 agencies.

13.2.2. AC Responsibilities:

13.2.2.1. Prior to commencing CSAR support, reference the On-scene Commander Checklist.

13.2.2.2. Notify Tactical Air Control Command and Control (TAC C2) of on-scene arrival and estimated endurance time.

13.2.2.3. Assume on-scene commander duties as required until relieved by TAC C2 or another aircraft. As other aircraft arrive, ACs will:

13.2.2.3.1. Establish contact on channels other than Guard.

13.2.2.3.2. Obtain aircraft type identification, endurance time and rescue capability. Assign an altimeter setting, altitude separation, radio frequencies, search areas and patterns.

13.2.2.4. Maintaining radio contact with the controlling agency and other supporting assets through coordination with the MCC. Monitoring assigned radios through coordination with the MCC, and pass relevant information.

13.2.2.5. Maintain a log of all turn times, directions and locations.

13.2.2.6. Coordinate with the MCC and increase speed as necessary to facilitate faster direction finding.

13.2.2.7. When requested, provide remaining time on station.

13.3. Communications with a Downed Aircrew. If tasked, establish direct communications with the downed aircrew as soon as possible. Accomplish this on the prescribed CSAR frequency IAW SPINS or the Evasion Plan of Action (EPA).

13.3.1. Communications procedures should instill confidence in the downed aircrew so they know professional assistance is at hand.

13.3.1.1. Coordinate immediate action items first and supplemental items as the mission progresses. Communications may be lost or the downed crew may be forced to discontinue communications.

13.3.1.2. Avoid long transmissions, but provide pertinent data at periodic intervals to reassure the downed crew that contact is being maintained. Make all messages clear, concise and use a confident tone.

13.3.1.3. If all efforts to contact the downed aircraft fail, transmit pertinent information and instructions in the blind as though the downed aircrew is receiving but unable to acknowledge.

13.3.2. Initial Communications Procedures.

13.3.2.1. Identify yourself and advise that you are enroute to support.

13.3.2.2. Instruct the downed crew to use the present primary frequency and not to break contact. Designate a secondary frequency.

13.3.2.3. Verify the nature of the emergency and the downed aircrew's intentions.

13.4. Departing Search Area.

13.4.1. When departing the search area at bingo fuel, appoint another aircraft as on-scene commander. If another search aircraft has not arrived, reconfirm the downed aircrew's position, and advise them when further assistance will arrive, if known.

13.4.2. Notify the TAC C2 agency and on-scene Search and Rescue (SAR) aircraft of off-station time including any unaccomplished tasks.

BRAIN S. LAIDLAW, Major General, USAF
Director of Operations

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

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Prescribed Forms

None

Adopted Forms

AF Form 4076, *Aircraft Dash 21 Equipment Inventory*

AF Form 651, *Hazardous Air Traffic Report (HATR)*

AF Form 664, *Aircraft Fuels/Ground Servicing Documentation Log*

AF Form 70, *Pilot's Flight Plan and Flight Log*

AF Form 711B, *USAF Mishap Report*

AF Form 8, *Certificate of Aircrew Qualification*

AFTO Form 46, *Prepositioned Aircrew Flight Equipment*

AFTO Form 781, *ARMS Aircrew/Mission Flight Data Document Aircraft Forms*

CBP Form 7507, *General Declaration (Outward/Inward) Agriculture, Customs, Immigration, and Public Health*

DAF Form 1297, *Temporary Issue Receipt*

DAF Form 457, *Hazard Report*

DAF Form 522, *Ground Weapons Training Data*

DAF Form 847, *Recommendation for Change of Product*

DD Form 1252, *US Customs and Border Protection (CBP) Declaration for Personal Property Shipments*

DD Form 1610, *Request and Authorization for TDY Travel of DoD Personnel*

DD Form 1801, *DoD International flight Plan*

DD Form 1896, *DoD Fuel Identaplate*

DD Form 1898, *Energy Sales Slip*

SF 44, *Purchase Order-Invoice-Voucher (Storage Safeguard Form)*

Abbreviations and Acronyms

AC—Aircraft Commander

ACC—Air Combat Command

ACDE—Aircrew Chemical Defense Ensemble

ACF—Acceptance Check Flight

ADVON—Advanced Echelon

AFE—Aircrew Flight Equipment

AFTO—Air Force Technical Order

AGE—Aircraft Ground Equipment

AGL—Above Ground Level

AIM—Aeronautical Information Manual

AIR Card—Aviation Into-Plane Reimbursement Card

ALEP—Aircrew Laser Eye Protection

AMT—Airborne Maintenance Technician

AOM—Airplane Operating Manual

AOR—Area of Operations

AP—Area Planning

APOD—Aerial Port of Debarkation

APU—Auxiliary Power Unit

ARMS—Aviation Resource Management

ARTCC—Air Route Traffic Control Center

ASOS—Automated Surface Observing System

ASRR—Airfield Suitability and Restrictions Report

ATC—Air Traffic Control

ATIS—Automated Terminal Information Service

ATO—Air Tasking Order

AVPOL—Aviation Petroleum Oil and Lubricants

AWOS—Automated Weather Observation System

BASH—Bird/Wildlife Aircraft Strike Hazard

BWC—Bird Watch Condition

C/JFACC—Combined/Joint Force Air Component Commander

C2—Command and Control

CBP—Customs and Border Protection

CBRNE—Chemical, Biological, Radiological, Nuclear and High Yield Explosives

CCMCS—COMPASS CALL Mission Crew Simulator

CCMD—Combatant Command

CFL—Critical Field Length

CFP—Computer Flight Plan

CG—Center of Gravity

CHOP—Change in Operational Control

CIRVIS—Communications Instructions for Reporting Vital Intelligence Sightings

CNDC—Canadian National Defense Contract

COMAFFOR—Commander Air Force Forces

COMSEC—Communications Security

CONUS—Continental United States

CPDLC—Controller-Pilot Data Link Communications

CRA—Country Risk Assessment

CRM—Cockpit/Crew Resource Management

CSAR—Combat Search and Rescue

CVR—Cockpit Voice Recorder

DAF—Department of the Air Force

DAFMAN—Department of the Air Force Manual

DDA—Derived Decision Altitude

DD—Department of Defense (abbreviation for Forms only)

DEPORD—Deployment Order

DER—Departure End of Runway

DETCO—Detachment Commander

DH—Decision Height

D-ICS—Digital Inter Communication System

DLA Energy—Defense Logistics Agency Energy

DoDI—Department of Defense Instruction

DO—Director of Operations (Squadron level)

DoDM—Department of Defense Manual

DOW—Department of War

DPT—Deployment Planning Team

DSN—Defense Switched Network

DSR—Deployed Status Report

ECG/CC—Electronic Combat Group Commander

ECG/EGV—Electronic Combat Group Standardization and Evaluation

ECG/SE—Electronic Combat Group Safety

ECG—Electronic Combat Group

EEBD—Emergency Escape Breathing Device

EMI—Electromagnetic Interference

EOD—Explosive Ordinance Disposal

EPA—Evasion Plan of Action

EP—Evaluator Pilot

EPOS—Emergency Passenger Oxygen System

ERO—Engines Running Onload/Offload

ETA—Estimated Time of Arrival

ETP—Equal Time Point

FAA—Federal Aviation Administration

FAF—Final Approach Fix

FAR—Federal Aviation Regulation

FAS—Fuels Automated System

FBO—Fixed Base Operation

FCF—Functional Check Flight

FCG—Foreign Clearance Guide

FCIF—Flight Crew Information File

FCM—Flight Control Manual

FDP—Flight Duty Period

FDR—Flight Data Recorder

FIH—Flight Information Handbook

FL—Flight Level

FLIP—Flight Information Publication

FMI—Frequency Modulation Immunity

FMS—Flight Management System

FOL—Forward Operating Location

FP—Basic Qualified Pilot

FRAG—FragmentationFSAF—First Suitable Airfield

FSS—Flight Service Station

ft—feet

GCCS—Global Command and Control System

GCE—Ground Crew Ensemble

GDSS2—Global Decision Support System 2

GNC—Global Navigational Chart

GNSSU—Global Navigation Satellite Systems

GOLD—Global Operational Datalink

GP—General Planning

GPS—Global Positioning System

GW—Gross Weight

HAA—Height Above Airport

HAT—Height Above Touchdown

HATh—Height Above Threshold

HATR—Hazardous Air Traffic Report

HF—High Frequency

ICAO—International Civil Aviation Organization

IFE—In-flight Emergency

IFR—Instrument Flight Rules

ILS—Instrument Landing System

IMC—Instrument Meteorological Conditions

IPE—Individual Protective Equipment

IP—Instructor Pilot

ISR—Intelligence, Surveillance and Reconnaissance

JBSA—Joint Base San Antonio

JMPS—Joint Mission Planning System

JSAM—Joint Service Aircrew Mask

lbs—Pounds

LPU—Life Preserver Unit

LRC—Long Range Cruise

LSAF—Last Suitable Airfield

MAJCOM/A3—Major Command Directorate of Operations

MAJCOM—Major Command

MCC—Mission Crew Commander

MC—Mission Commander

MDA—Minimum Descent Altitude

MDS—Mission Design Series

ME—Mission Essential

MEP—Mission Essential Personnel

MESL—Minimum Essential Subsystem List

mic—Microphone

ML—Misson Lead

MMEL—Master Minimum Equipment List

MOPP—Mission-Oriented Protective Posture

MPC—Mission Planning Cell

MRE—Meal Ready to Eat

MSC—Multi Service Corporation

MSL—Mean Sea Level

NAF/A3V—Numbered Air Force Standardization and Evaluations

NAVAIDS—Navigational Aids

NMAC—Near Midair Collision

NM—Nautical Mile(s)

NORAD—North American Aerospace Defense Command

NOTAM—Notice to Air Missions

NSN—National Stock Number

NVD—Night Vision Device

OCF—Operational Check Flight

OCONUS—Outside the Continental United States

OEI—One Engine Inoperative

OE—Observer Electronic Warfare Officer

OPCON—Operational Control

OPLAN—Operations Plan

OP—Observer Pilot

OPORD—Operations Order

OPR—Office of Primary Responsibility

OSS—Operational Support Squadron

OSW—Office of the Secretary of War

OWS—Operational Weather Squadron

PAM—Pyridine Aldoxime Methel

PBN—Performance Based Navigation

PFD—Primary Flight Display

PF—Pilot Flying

PIC—Pilot in Command

PM—Pilot Monitoring

POL—Petroleum, Oil and Lubricants

psi—Pounds per Square Inch

RAG—Runway Arresting Gear

RAIM—Receiver Autonomous Integrity Monitoring

RA—Resolution Advisory

RAS—Runway Arresting System

RCR—Runway Condition Reading

RNP—Required Navigation Performance

ROE—Rules of Engagement

RRFL—Required Ramp Fuel Load

RSC—Runway Surface Condition

RVR—Runway Visual Range

SAR—Search and Rescue

SF—Standard Form

SIGMET—Significant Meteorological Information

SIIs—Special Interest Items

SITCO—Shell™ International Trading Company

SITREPS—Situation Reports

SM—Statute Miles

SOC—Squadron Operations Center

SPINS—Special Instructions

SSO—Special Securities Officer

Stan/Eval—Standardization and Evaluation

T.O.—Technical Order

TAC C2—Tactical Air Control Command and Control

TCAS—Traffic Alert and Collision Avoidance System

TDY—Temporary Duty

TEMPO—Temporary Group

TERPS—Terminal Instrument Procedures

TOLD—Takeoff Landing Data

TX—Texas

UHF—Ultra-High Frequency

USAF—United States Air Force

VCSL—Voice Call Sign Listing

VFR—Visual Flight Rules

VHF—Very High Frequency

VMC—Visual Meteorological Conditions

WEZ—Weapons Engagement Zone

WPS—Weapons Squadron

Office Symbols

16 AF/A3V—16th Air Force Standardization and Evaluation

55 ECG/EGV—55th Electronic Group Standardization and Evaluation

ACC/A3—Air Combat Command Director of Operations

ACC/A3C—Air Combat Command, Command and Control, Intelligence, Surveillance, and Reconnaissance Operations Division

ACC/A3CR—Air Combat Command Airborne Reconnaissance Operations Branch

ACC/A3T—Air Combat Command Flight Operations Division

ACC/A3TO—Air Combat Command Weapons and Tactics Branch

ACC/A3TV—Air Combat Command Standardization and Evaluations Branch

AF/A3T—Air Force Training and Readiness Directorate

AF/ACTF—Air Force Aircrew Task Force

AFFSA—Air Force Flight Safety Agency

AFSC/SEF—Air Force Safety Center Aviation Center Flight Safety Division

AMC/A3AS—Air Mobility Command Airfield Sustainability Branch

Attachment 2

TRAINING POLICY AND RESTRICTIONS

A2.1. Touch-and-Go Landings:

A2.1.1. Touch-and-go landings are authorized under the following conditions:

A2.1.1.1. Flown under direct IP supervision.

A2.1.1.2. Minimum runway length of 8,000 feet or the greater of Runway Arresting System (RAS)/ Runway Arresting Gear (RAG) distances, whichever is greater.

A2.1.1.3. Crosswind component must be within the recommended zone of the landing crosswind chart.

A2.1.1.4. Authorized when normal wake turbulence criteria (IAW AFMAN 11-202V3) are met.

A2.1.1.5. Comply with all flight manual restrictions and procedures to include performance degradation with fuel, cargo limits, etc.

A2.2. Stop-and-Go Landings:

A2.2.1. Stop-and-go landings are authorized only on designated training, evaluation, or currency missions and may be performed by any qualified EA-37B pilot.

A2.2.2. When stop-and-go landings are performed, the runway remaining for takeoff must be equal to or greater than Reference Accelerate-Go Distance or Reference Accelerate-Stop Distance, whichever is greater. Crosswind component must be in the recommended zone of the landing crosswind chart. Ceiling and visibility must be at least 300 feet and 3/4 mile (RVR 4000).

A2.2.3. Do not perform stop-and-go landings in conjunction with no-flap landings, when normal wake turbulence criterion is not met, or when intercepting/crossing the flight path of a heavy jet while performing an approach or landing.

A2.3. Simulator-Only Maneuvers. Do not perform the following maneuvers or procedures in flight:

A2.3.1. Full stalls

A2.3.2. Rudder force reversals or fin stalls

A2.3.3. Spins

A2.3.4. Simulated runaway trim malfunctions

A2.3.5. Simulated single engine approach or landing

A2.3.6. Simulated single engine takeoffs

A2.3.7. Practice Aborted Normal Takeoff.

A2.4. Simulated Instrument Flight. Do not use a hood or other artificial vision-restricting device for any phase of flight. Simulated instrument flight may be flown and logged without the use of such devices.

A2.5. Actual Engine Shutdown and Airstart. For Functional Check Flight usage only. Direct IP supervision is required; ECG/CC approval is required. Engines may be shut down during daylight, in VMC, and at altitudes no lower than 2,500 feet AGL.

A2.6. Go-Around and Missed Approach. Initiate a planned go-around/missed approach no lower than:

A2.6.1. Precision approach – Decision Height (DH)/Decision Altitude (DA) (or 200-feet HAT, whichever is higher).

A2.6.2. Non-precision approach – Minimum descent altitude depicted on approach plate.

A2.6.3. Visual Approach – 200-feet AGL for simulated emergencies (no minimum for non-emergency).

A2.6.4. VFR pattern or circling/sidestep approach (after circling maneuver has begun) – 200 feet AGL for simulated emergencies (no minimum for non-emergency).

A2.6.5. Restricted Low Approach (aircraft, equipment, or personnel are on the runway) – 500 feet AGL.

A2.7. Partial Flap Approach and Landing. Direct IP supervision is required.

A2.7.1. Restrictions:

A2.7.2. . Maximum gross weight is 70,000 pounds.

A2.7.3. . Authorized in day IMC with ceilings and visibility at or above circling minimums.

A2.7.4. . Authorized at night with weather at or above 1,000-foot ceiling and 2SM visibility or circling minimums whichever is higher.

A2.7.5. . Use 20 flaps for a go-around from a partial flap approach.